Entry Timing with Regulatory Frictions and Fighting Brands: Evidence from Generics and Authorized Generics*

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Abstract

Generics face regulatory frictions while attempting to enter a market, and the branded drug manufacturer can release an additional product called an Authorized Generic (AG) at any time to compete with generic entrants. Using sales and revenue data on the US, we estimate a structural model of generic entry and timing of AG release, then simulate counterfactuals to study market dynamics in this setting. Generics and AGs deter each other differently – generics can apply for entry earlier, but AGs can launch earlier. Faster generic approval leads to weakly fewer generic entrants, lower prices in the early periods of the market, and weakly higher prices in later periods. A ban on AG leads to weakly more generics, and often multiple new generics. The effect on prices is mixed - while there is increased price competition from more generics in the later stages of the market, it is counteracted by lack of an additional competitor through the AG in the early stages of the market.

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1 Introduction

We study market dynamics when potential entrants face regulatory frictions during entry, and the incumbent can launch additional products when facing new entry. Specifically, we look at generic drug manufacturers' entry decisions into molecule-formulations. Generics need to get approval from Food and Drug Administration (FDA) to launch; however, the time to get approval is uncertain and lengthy. In addition, they face strategic responses from the branded drug manufacturer that deter or blunt generic competition. One such strategy is for the brand manufacturer to release an Authorized Generic (AG). AGs are chemically identical to the branded drug but without the brand label attached. These are priced substantially lower than the brand and close to generics' prices. This (potential) additional competition from a similar product worsens the prospects of generics considering entry into the market. In the marketing literature this is known as a "fighting brand strategy".[1]

In this paper, we shed light on three issues. First, we study the economics of entry decisions when potential entrants face both uncertain approval times and fighting-brand launches. Second, we examine the effect of reduced regulatory frictions on market outcomes. Third, we illustrate the impact of AGs on short-term and long-term market outcomes, including generic entry deterrence. This can inform policy discussions about encouraging versus discouraging such fighting brand strategies.

To do so, we develop a stylized structural model of i) generics making static entry decisions, and ii) the branded incumbent making a dynamic timing decision to release an AG. In this setting, regulatory frictions manifest as generics receiving FDA approval stochastically over time, rather than being able to launch immediately. We perform counterfactual simulations to show the following. First we study how generics and AGs can deter each other. Because generics move first, they can apply for entry earlier; however, since AGs do not need regulatory approval, they can deter generics by being able to launch their products into the market earlier. Second, a faster approval rate leads to weakly *fewer* generic entrants, lower prices in the early periods of the market, and weakly higher prices in later periods of

[1] When facing new entrants, incumbents sometimes engage in price discrimination through release of fighting brands. In a fighting brand strategy, the incumbent releases a low brand-value version of its existing product, called a fighting brand. The high brand-value product charges high price and the fighting brand charges low price. This segments the market, with fighting brand competing with new entrants and original product serving the higher end of the market. An examples of fighting brands (from Bourreau et al. (2021)) is Intel with the Pentium series (brand) and Celeron (fighter brand) to compete with AMD.

the market. Finally, we show that a ban on AG leads to weakly more generic entrants, and sometimes multiple additional generics. The AG ban generally leads to higher prices in the early periods of the market. Prices are lower in the later periods of the market if the AG would have deterred multiple generics. The takeaway from simulating the impact of an AG ban is that while there is increased price competition from weakly more generic entrants in the later stages of the market, it is counteracted by lack of additional price competition from the AG in the early stages of the market.

The setting and our contributions are useful for several reasons. First, since generic competition is a key driver in lowering drug prices and spending, understanding the economic mechanisms at play is important for crafting policies. Second, while our structural model is strictly about AGs, we believe it can be extended to study other strategic reactions against generics by branded drug manufacturers (product line extensions, REMS requirements, orphan drug classification, etc.) and specific regulatory frictions. In particular, it could shed light on biosimilars and biologics – biosimilars face substantial regulatory frictions, and there is evidence suggesting that biologic manufacturers are considering releasing Authorized Biologics to counter biosimilar entry. Finally, fighting brands are widely observed in many industries besides pharmaceuticals, so this study adds to our understanding of such marketing strategies, particularly entry deterrence motives.

An outline of our stylized structural model is as follows. The demand side is a discrete choice model where consumers first choose a group - brand, nonbrand (generics and AG), or outside option - then choose a product within the group. The supply side of the industry is modeled as two separate stages. In the first stage, generic manufacturers make a static entry decision on whether to enter a molecule-formulation market. In the second stage, a dynamic game begins where every period, generics who decided to enter are randomly approved for product launch by the FDA, the branded drug manufacturer decides whether to release an AG, and market participants compete on prices. This two-stage model setup is because unlike generics, AGs do not need FDA approval to launch. We solve the two-stage supply model by backward induction, and as a result allow for AG and generics to form expectations about each others' entry and pricing decisions when making a choice.

We use data on national sales and revenue for prescription drugs in the US between 2004-2016 (IQVIA National Sales Perspective). We use this to estimate demand and supply side parameters. Our empirical measure of regulatory frictions is estimating the average generic entry rates for every quarter since loss-of-exclusivity; these are used by generics to form expectations about entry dates. Prices in different industry states are predicted using a reduced-form regression to avoid making pricing conduct assumptions. Following previously published papers, our market definition is the molecule-formulation pair (Conti and Berndt (2020), Wang et al. (2022), Yurukoglu et al. (2017)).

For our counterfactuals, we solve the model for different observed market specifications. An important element of model dynamics is that the early quarters of a market are most profitable for generics and AGs – there are few generics present as the rest wait to get FDA approval, resulting in lower price competition and high markups.

Our first set of counterfactuals illustrate how both generics and AGs can deter each other's entry, but in different ways. Generics can deter AG because they move first, i.e. they apply for entry and incur the entry cost first. However, AGs can deter generics because they can launch their product earlier than generics – the early launch significantly reduces generics' expected profits by providing price competition in the early profitable periods of the market. To showcase these two forces, we first show that no AG is released when the generics can launch immediately and the AG has no cost or demand advantage; we then show that AG is able to deter more generics when it can launch earlier rather than being forced to delay.

Our second set of counterfactuals study how reducing regulatory frictions affects market outcomes. Faster generic approval has been a priority of the FDA since the 2010s (Berndt et al. (2018)). We find that faster approval rates leads to weakly fewer generic entrants, lower prices in the early periods of the market, weakly higher prices in the later periods of the market, and that the effect is more muted if AGs may be present in the market. Why does a faster generic approval rate lead to lower generic entry? While a faster approval rate means a longer stream of revenue accruing to a generic from earlier launch, it is counteracted by greater presence of rivals upon launch. The latter means greater price competition and lower profits upon launch. Greater competition upon launch offsets the advantage from an early launch, reducing lifetime generic profits and therefore weakly reducing generic entry.

Our third set of counterfactuals study how the presence of AGs affects short-term and long-term market outcomes compared to an absence of AGs. This can inform policymakers about whether AGs should be encouraged or banned. We show that a ban on AG leads to usually leads to more generic entrants, and in fact may lead to multiple additional generics entering the market. The AG ban can incentivize multiple new generic entrants because it is a more potent competitor than independent generics as a result of its early-launch advantage, and so has a more negative impact on generics' lifetime expected profits. If the AG is banned, not only is there one fewer competitor, but it is also one less competitor in the most profitable periods in the market.

The effect on prices from the AG ban is mixed. When an AG is released, it is usually in the early periods of the market just after loss-of-exclusivity. This means that generics entering the market early will face an additional competitor through the AG. Therefore, the AG keeps prices low in the early periods of the market.[2] However, if the AG deters multiple generics from entering, then prices in the later periods of the market would be higher with the AG due to missing generic competition.

Contributions and related literature: This paper contributes to the following strands of literature. First, we contribute to a small literature on Authorized Generics. We are the first to study the impact of AGs on generic firms by using a structural model of entry. Moreover, our model incorporates this interaction in a rational expectations framework. This allows us to trace out feedback between AG and generic decisions, and in particular allows us to study how the presence/absence of AG affects generics' decisions. Furthermore, our model allows us to explore a wider variety of economic effects from the presence of the AG. A few papers have used reduced-form evidence to study Authorized Generics. Notably, Appelt (2015) uses a recursive bivariate probit regression to show that AG entry does not impact generic entry in Germany. An important source for us is a report from the Federal Trade Commission (FTC (2011)) that leveraged many information sources to explain the decision to release the AG and generic manufacturers' reactions to it. FTC (2011) mostly focuses on how AG release may disincentivize Paragraph IV lawsuits by generics and does so by compiling industry data and company documents; this paper instead focuses on how AGs and generics interact when a Paragraph IV lawsuit has not been filed. Both FTC (2011) and this paper finds that the majority of AGs are released in the absence of a Paragraph IV lawsuit. A recent paper by Fowler et al. (2023) uses industry data to

[2]One exception is markets where the AG would have chosen to enter later. In such markets, the AG ban has minimal effect on early-period prices since AG is absent during that period, and in fact may even lower prices if an additional generic receives early FDA approval.

describe broad trends about Authorized Generics over 2010-2019, most of which matches our findings; see Appendix F for a comparison.

Second, we add to the literature on generic drug entry, particularly generic entry deterrence. While many papers have studied generic entry in the US, few use structural methods to model such decisions. Doing so allows us to see how changing key economic parameters affects entry incentives by generics. Furthermore, we contribute to the literature developing a structural model of generic entry deterrence that may be used to study other generic-deterrence strategies. An important paper for our purposes is Ching (2010), who studies generic entry and brand's dynamic pricing in 1984-1990 to model learning dynamics. We adopt part of our entry model from this paper. Other notable papers are Gallant et al. (2017), Morton (1999), Reiffen and Ward (2005), and Starc and Wollmann (2023). A paper that we think of as closely complementing ours is Wang et al. (2022), which uses a static entry model and reduced-form profit functions to understand the factors that promote generic entry into a molecule-formulation.[3]

Third, we contribute to a very sparse Empirical IO literature on fighting brands. We are one of the very few papers to build and estimate a model of an incumbent releasing a fighting brand. There is a theoretical literature on fighting brands, notably Anderson and Dana (2009), Deneckere and McAfee (1996), Johnson and Myatt (2003), and Nocke and Schutz (2018). An important empirical paper studying fighting brands is Bourreau et al. (2021). They show that in the French mobile telecommunications market, releasing fighting brands occurred due to a breakdown of collusion, and use a structural model of demand and supply to study this phenomenon.

[3] Our papers each focus on one aspect of the industry while abstracting away from another. In contrast to our paper, Wang et al. (2022) capture heterogeneity on the generics side, look at more molecule-formulations, and consider markets where generics never entered. In particular, they investigate how firm-specific covariates affect entry decisions, while our paper instead assumes that generics are homogeneous. On the other hand, unlike our paper, they i) do not explicitly account for dynamics, ii) do not allow for authorized generics, and iii) do not estimate a demand system and price competition (instead using a reduced-form expression to proxy for per-period profits). Our paper can track how market shares and prices change as the competitive landscape or policy environment changes, and use it to illustrate the knock-on effect on entry behavior. Furthermore, by accounting for dynamics, we can better simulate the impact of dynamic changes like faster FDA approvals and bans. As a result, we believe we are better positioned to answer how the number of generic entrants change due to faster FDA approval rates, something Wang et al. (2022) study, and we reach the opposite conclusion to their paper. In short, Wang et al. (2022) is better for understanding how firm-specific heterogeneity affects generic entry decisions into a wide range of markets, while our paper is better for thinking about market dynamics and authorized generics.

Fourth, we construct a parsimonious structural model of pharmaceutical product entry that is quick to solve and can be easily extended in various directions (such as consumer heterogeneity, firm heterogeneity, presence of intermediaries, etc.) in future research. In doing so, we contribute to an extensive literature on static entry games and dynamic single-agent and oligopoly games. For static entry, key references include Berry (1992), Berry et al. (2016), Bresnahan and Reiss (1991), Mazzeo (2002), and Seim (2006). For dynamic single-agent and oligopoly games, some important references are Aguirregabiria and Mira (2007), Benkard (2004), Collard-Wexler (2013), Gowrisankaran and Rysman (2012), Igami (2017), Igami (2018), Igami and Uetake (2020), Ericson and Pakes (1995), Pakes et al. (2007), Rust (1987), and Ryan (2012); see Aguirregabiria et al. (2021) for a detailed overview of the literature. We adopt part of our model and estimation method from Igami (2018).

Fifth, we contribute to a somewhat limited empirical literature on entry deterrence and preemption. Unlike most papers in this area, we tackle empirical issues regarding entry deterrence differently. The fundamental empirical problem is that when a potential entrant does not enter, the researcher cannot tell if this was due to deterrence or due to absence of entry threat to begin with (see Cookson (2017) and Gil et al. (2024) for a detailed discussion and creative workarounds). In our paper, we tackle this issue by building a model of entrant's choices, and using it to predict entry decisions in different markets. The sequential nature of moves in this setting (i.e. generics make entry decision first followed by AG) leads to a clean solution of the entry deterrence problem. Empirical papers on this topic include Cookson (2017), Fang and Yang (2024a), Fang and Yang (2024b), Gil et al. (2024), Goolsbee and Syverson (2008), Hünermund et al. (2014), Igami and Yang (2016), Schmidt-Dengler (2024), and Seamans (2012). There is an extensive theoretical literature on this topic, which can be found in the references above.

Finally, our paper relates to a large literature on pharmaceuticals. A vast amount of work has been done on the theoretical and empirical side of this industry. Frank and Salkever (1992) was the first to lay out a theoretical model for why branded drug prices often stayed above generic prices. A non-exhaustive list of references that informed the current paper include Arcidiacono et al. (2013), Berndt et al. (2007), Bhattacharya and Vogt (2003), Bokhari and Fournier (2013), Bokhari et al. (2020), Chen et al. (2022), Dubois et al. (2022), Dubois and Majewska (2022), Ellison and Ellison (2011), Frank and Salkever (1997), Ganapati and McKibbin (2023), Grennan et al. (2021), Hermosilla and Ching (2023), Lakdawalla (2018), Olson and Wendling (2018), Olssen and Demirer (2199), Reiffen and Ward (2007), Tenn and Wendling (2014), and Tunçel (2024); see Morton and Kyle (2011) for an overview. More specifically, we contribute to the literature of using structural models to understand the economics of the pharmaceutical industry; see Ching et al. (2019) for an overview. Berndt et al. (2017), Berndt et al. (2018), and Conti and Berndt (2020) provide important descriptive evidence about the aggregate dynamics and inner workings of the industry which we use to motivate our model.

2 Institutional Background

After the branded drug manufacturer loses market exclusivity, generics enter the market by applying for FDA approval. However, gaining FDA approval is lengthy, costly, and takes an uncertain amount of time. The brand drug manufacturer can respond by releasing an authorized generic (AG). The AG is chemically identical to the brand drug but without the brand label attached. Unlike generics, the AG can be launched at any time without having to wait for FDA approval. The general patterns after loss of exclusivity are: branded drug prices stay elevated, generics and AG charges low price which further declines with more competition, and the branded drug loses most of its market share to generics and AGs over time.

2.1 Generic Drugs and FDA approval

A pharmaceutical manufacturer invests in R&D to discover a drug, then patents the molecule and conducts tests for efficacy and safety. This drug is commonly referred to as the "branded drug" (see Dubois et al. (2015) and Khmelnitskaya (2023), and the references therein). Throughout we refer to the manufacturer that pioneered the molecule as the "branded drug manufacturer" and the pioneer molecule as the "branded drug". We use the term "nonbrand" to indicate both generics and AGs.

After loss of exclusivity, generic drugs apply for entry into the molecule-formulation to the FDA. The FDA's approval process is costly for the generic manufacturer, lengthy, and has uncertain duration.

To gain FDA approval, generic manufacturers need to prove to the FDA that their products are "bioequivalent" to the branded drug and can be produced at scale. Bioequivalence is defined to mean "product has the same active ingredient, dosage form, strength, route of administration and conditions of use" as the reference branded drug.[4] Generic manufacturers have to give further evidence that they can manufacture the drug accurately, at scale for commercial distribution, and according to Good Manufacturing Practices.[5] This might further involve a back and forth between the FDA and generic applicant until the FDA is satisfied with the production line.[6] An important aspect of proving bioequivalence is that the generic firm needs to reverse-engineer the branded drug. Since some aspects of the manufacturing process for the branded drug may be undocumented or kept as trade secrets by the branded manufacturer, generic firms must experiment with production to meet regulatory requirements. Estimates of the cost of ANDA application range from $3 million to $15 million.

The generic manufacturer then needs to submit an Abbreviated New Drug Application (ANDA) with all the relevant information attached. The mean approval time for an ANDA is between 32-40 months for our data period. Approval times of generics vary due to the amount of work involved in verifying the evidence submitted in support of the entrant's ANDA, as well as the FDA's workload. In the past several decades, the FDA has experienced a significant backlog of pending ANDA applications. Policies such as the GDUFA – and its reauthorization every 5 years – have given the FDA more resources to reduce the time and increase the scope of generic drug approvals (Berndt et al. (2018), Wang et al. (2022)).

In this paper, we refer to applying for FDA approval as "applying for entry", and refer to generics launching their product after FDA approval as "entry". Thus, generics *apply to enter* a market by applying for FDA approval, and *enter* the market after receiving FDA approval.

[4] Source: https://www.fda.gov/media/71401/download

[5] See information from the FDA on Good Manufacturing Practices: www.fda.gov/drugs/pharmaceutical-quality-resources/current-good-manufacturing-practice-cgmp-regulations.

[6] Bioequivalence ensures that generic drugs are not lower quality compared to their equivalent branded drugs - by definition, their bioequivalence means that they can be substituted with the branded product and have the same effect. Thus, generic drugs compete with their branded equivalents in the US on the basis of their availability and price.

2.2 Authorized Generics

In response to generic entry, the branded drug manufacturer can release an Authorized Generic (AG). The AG is chemically identical to the branded drug in terms of molecule structure and formulation, except that it does not have the brand label attached.

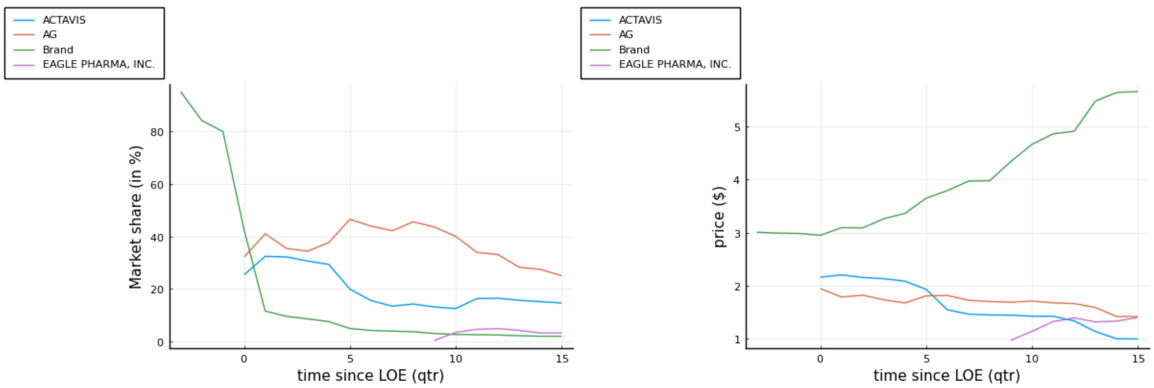
A key detail about product launch in this setting is that, unlike generics, AGs can be introduced anytime and without FDA approval. This is because they are riding on the branded drug's approval from the FDA.

Why would a branded drug manufacturer not release an AG whenever it faces generic entry? There are several reasons. First, the AG could cannibalize the branded drug's sales. Second, the AG would lead to generics lowering their prices in response, further increasing the price gap between brand and generics products; this causes branded drug sales to fall. Third, it could be due to supply side reasons such as manufacturing delays and costs. Fourth, it could be that many generics got FDA approval in the very first period after loss-of-exclusivity, meaning that the AG will make little profit. In our model we account for all these factors.[7]

Finally, a brand manufacturer does not always manufacture and market its own AG. It may be done by a generic subsidiary of the brand. It may also be done by firms which specialize in releasing AGs on behalf of other manufacturers, most notably Greenstone and Prasco. Finally a generic manufacturer may team up with a brand manufacturer to release its AG. In these cases, the non-brand firm takes over part of the marketing and/or manufacturing responsibilities, and in return gets a share of the profits. See Section G for more details on this from FTC (2011).

[7] This raises the question of why the brand manufacturer does not simply lower the price of the branded product instead of releasing the AG. First, there is a price discrimination motive where consumers who like the branded drug are willing to pay a high price for it, so having the expensive brand and inexpensive AG in the market allows the manufacturer to capture both types of consumers. Second, if the generic entrant has to leave the market (e.g. due to legal reasons, patenting issues, technical or manufacturing problems) the brand may not be able to raise its prices back up without facing backlash from buyers. Third, if the brand product's price is lowered, it is hard to justify high prices on line extensions. All the points mentioned above are corroborated in FTC (2011).

Figure 1: Evolution of market shares (left) and prices (right) for Diclofenac-Misoprostol (oral formulation).



2.3 Evolution of Market Shares and Prices

In terms of market prices and shares, the general pattern we see in the data is as follows. After loss of market exclusivity by the branded drug, generics enter the market and undercut the brand's price. The branded drug's price typically remains stable or increases. An AG is sometimes released by a branded drug manufacturer in response to generic entry. The AG and generics price their products close to each other. Over time, the branded drug's market share dramatically falls, while those of nonbrand drugs rises. Examples of this pattern can be seen in Figure 1 for a molecule-formulation.

3 Data and Descriptive Statistics

We leverage data on quarterly sales and revenue of prescription drugs in the US from 2004-2016, and use this to construct market shares and prices. We define a market to be a molecule-formulation; and of the 241 molecule-formulations we restrict our attention to, about 45% see an AG released. Most AGs are released soon after the first generic entry occurs. Most markets have between 1-11 competitors.

3.1 Data

The data for this paper comes from IQVIA's National Sales Perspective™(NSP) database and covers sales in US for 2004 - 2016. NSP™ data contains national sales of prescription drugs to pharmacies, clinics, hospitals, and other distribution channels, aggregated to the quarterly level. Each observation includes the name of the molecule (active ingredient) and branded name (if relevant), quarter, sales amount in drug unit volume, sales amount in U.S. dollars, supplier of the product, distribution channel, product form (e.g., oral, injectable), and other information including therapeutic class, product launch date, etc. Following previous literature (Conti and Berndt (2020), Wang et al. (2022), Yurukoglu et al. (2017)), we refer to a drug as a group of products having the same molecule (or active ingredient) and product form.

For our purposes, a drug's therapeutic class describes what broad diagnoses it targets; a drug's molecule describes the active ingredients present; and a drug's formulation is its method of delivery, e.g. oral, injectable, etc.

We now give a brief overview of the NSP™ database; this closely follows Berndt et al. (2017), which contains a more detailed description of the entire dataset.

The database is constructed through a projected audit that tracks drug sales from manufacturers or wholesalers to pharmacies and other distribution outlets. The measure of "revenue" for each drug is the total amount paid for the purchase of a product to the manufacturer.[8] Furthermore, the NSP database reports volumes of each drug sold in "standard units", where a standard unit is defined as the smallest unit of the drug standardized across dosage and therapeutic form (Dubois and Lasio (2018)). We compute prices as the ratio of total revenue to total quantity in standard units, following Dubois et al. (2022) among others. In addition, data on Authorized Generics and Paragraph IV lawsuits were hand-

[8]Notably, this measure of revenue does not include rebates that drug manufacturers give back to other intermediaries. Generic manufacturers do not pay rebates, but branded drugs do, so this creates potential measurement error for branded drug prices. To address this, we rerun our demand estimation allowing branded drugs to be charging 30% rebates (following the findings in Kakani et al. (2020) and Kakani et al. (2022)), and obtain very similar results.

While rebates could be significant before loss of exclusivity, the size of the branded drug's rebate *after* loss of exclusivity is unknown. A lot of the theoretical and empirical literature on pharmaceuticals state that even after controlling for rebates, branded drug prices are significantly higher than generic prices. This fact is also used to explain AG release in FTC (2011).

collected.[9] AGs were identified from FDA's maintained list of AGs, an online source maintained by some producers of AGs, and verifying with news reports on AG release. The list of molecules facing Paragraph IV lawsuits were collected from a publicly available list maintained by the FDA.[10]

To summarize, for each drug product we see quarterly sales in the US, revenue generated, list of active ingredients, formulation of the drug (oral, injectable, or other), and therapeutic class (ATC3). Sales for a drug-molecule-formulation are aggregated by dosage and strength. Finally, we define a market at the molecule-formulation level.[11]

3.2 Descriptive Statistics

While our dataset has comprehensive coverage across all prescription drugs sold in US, we look at a smaller subset of drugs for our project. We select molecule-formulations that saw its first generic entrant after 2004 and before 2016, for which we can identify presence or absence of AG with certainty, and those that are not surrounded by strong external circumstances (e.g. media outrage, serious or repeated lawsuits). This leaves us with 241 molecule-formulations. [12]

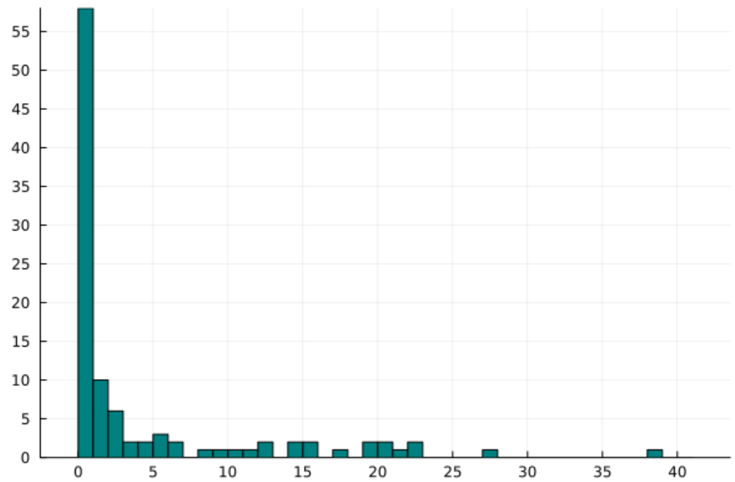
[9]We searched news stories about every molecule online to try to ensure there were no non-economic events surrounding them that would affect generic entry. Molecules which faced unnatural events such as lawsuits being settled then revived, facing new lawsuits after entry and being forced to discontinue, facing public pressure such as Epipen, etc. are left out of the final sample.

[10]The FDA list on AGs can be found here: https://www.fda.gov/media/77725/download.
The online source by AG producers can be found here: https://www.authorizedgenerics.com/product-finder/
The FDA list on Paragraph IV lawsuits can be found here: https://www.fda.gov/media/133240/download.

[11]This raises the question of why we define a market at the molecule-formulation level instead of a higher level, such as the therapeutic class. First, we include a molecule-formulation fixed effect in the demand function to account for presence of other molecule-formulations in the same therapeutic class. Second, similar market definitions have been used in various antitrust proceedings and empirical papers (Conti and Berndt (2020), Starc and Wollmann (2023), Wang et al. (2022), Yurukoglu et al. (2017)). Third, the fewer number of products allows us to capture substitution patterns between these bioequivalent products more accurately. Fourth, the branded drug's patent and generic approval holds at molecule-formulation level. Fifth, one drug can be in multiple therapeutic classes which makes modeling therapeutic classes as markets unclear. Finally, there is evidence suggesting that while there can be substitution across molecule-formulations within the same therapeutic class, usually the substitution is from a molecule-formulation without a generic to a molecule-formulation with one (Arcidiacono et al. (2013)). In this data we only look at periods after generic entry, so it is unlikely that people are switching away from the molecule-formulation we are studying.

[12]We now discuss how this sample selection ties in with our research question and interpretation of results. For studying AG entry, we believe this way of identifying markets AG may enter is quite realistic.

Figure 2: Histogram of time-difference between first generic entry and AG release period.



This figure is limited to 22 quarters, while in the dataset 4 AGs get released later than 22 quarters. This distribution has a 10th percentile of 0, 25th percentile of 0, median of 0, 75th percentile of 4.25, and 90th percentile of 16.4. See the Appendix for the exact distribution.

Table 1 shows descriptive statistics about our selected molecule-formulations: of the 241 molecule-formulations, about 45% see an authorized generic released, and over 67% are oral formulations. Note that each molecule-formulation has one branded drug and can have at most one AG.

For a molecule-formulation, we define the period of loss-of-exclusivity (LOE) to be the earliest quarter that saw positive generic drug sales.

Figure 2 shows when the AGs were released in these markets relative to generic entry. This is a histogram that shows how many quarters after first-generic-entry is AG released.

AGs are always only released in response to generic entry. In fact, FTC (2011) show company documents where, when outsourcing AG manufacturing, brand manufacturer imposes strong restrictions on the partner AG manufacturers to prevent them from launching the AG before the first generic has entered. Therefore, identifying markets with at least one generic as markets where the AG may enter is realistic, and so our AG cost estimation should be unaffected.

One type of market we do rule out with this is a market where no generic enters, but would have entered had it not been for the threat of AG release. Another type of market we rule out are those which are so unprofitable that a generic never enters. Most important prescription drugs see generic entry, so it is unclear how much we lose from not considering those markets. Finally, it isn't clear that these would affect our demand estimation or AG cost estimation.

Table 1: Descriptive statistics

	No. of molecule-formulations
Total	241
AG released	109
Formulation: Oral	162
Formulation: Injectable	58
Formulation: All Others	21
Faced Paragraph IV lawsuit	62

58 markets see AG release immediately with the first generic entry, 10 markets see AG release one quarter after the first generic entry, and 6 markets see AG released two quarters later. AGs continue being released sporadically after the third quarter, with the latest release happening 41 quarters after loss-of-exclusivity.

Next we look at the competitive landscape in our selected markets, specifically regarding generic presence. Most markets have between 1-11 generic competitors, with the maximum number of generics in a molecule-formulation being 19 in our sample. See Figure 5 in the Appendix for the distribution of total generics that enter into a molecule-formulation.

Table 15 in the Appendix has additional descriptive statistics on within-molecule-formulation market shares and revenues, and we summarize some of them here. A generic in a given quarter has a median within-molecule-formulation market share of 18%, while an AG has 22.04%. We also calculate a conservative estimate of lifetime revenue of a product by adding up all the revenue it generated within our dataset range; this is conservative becaues generics and AGs can continue selling their products after our final available data period. The median AG product has a lifetime revenue of $54.45 million, the 75th percentile is $201.78 million, and the 90th percentile is $444.5 million. More details can be found in Table 15. This further shows the importance of AGs in the prescription pharmaceuticals industry and understanding their market impact.

Next we study the competitive effects of generic entry on market shares, revenue, and prices; the results are reported in Table 2. Two patterns are particularly notable here. First, the first few generics have a large impact on the brand's market share and revenue; generics that enter later have a much smaller impact. This suggests that most switching from brand to nonbrand products happen with the first few generics. Second, the brand's price stays the same or rises as more generics come in. This pattern has been documented previously in the literature.

Table 2

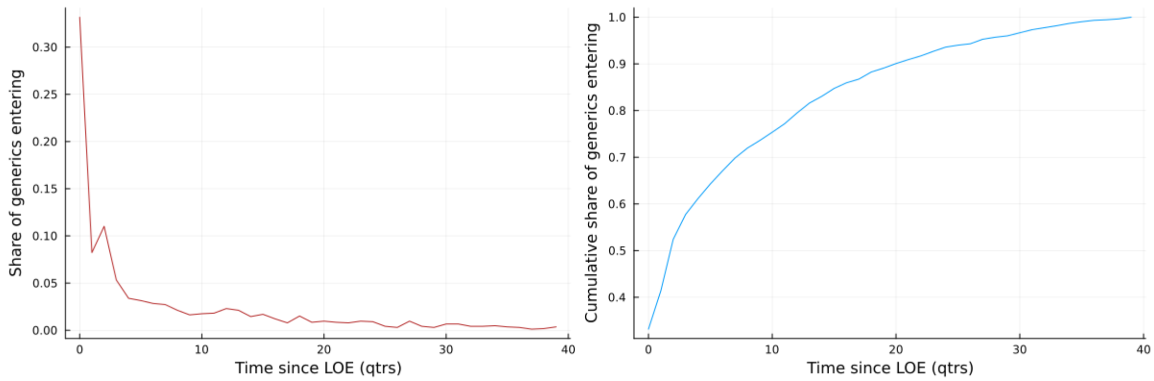
	log(share)	log(price)	log(revenue)
Brand	1.590***	0.291***	1.938***
	(0.080)	(0.029)	(0.080)
Brand x qtrs since LOE	-0.057***	0.007***	-0.059***
	(0.003)	(0.001)	(0.003)
Authorized generic × Count of generics	-0.133***	-0.143***	-0.278***
	(0.013)	(0.005)	(0.013)
Generic × Count of generics	-0.216***	-0.138***	-0.358***
	(0.008)	(0.003)	(0.008)
Brand × 1 (if count of generics <= 2) x Count of generics	-0.569***	0.038**	-0.502***
	(0.038)	(0.014)	(0.038)
Brand × 1 (if count of generics > 2) x (Count of generics - 2)	-0.103***	0.100***	-0.007
	(0.015)	(0.006)	(0.015)
Generic × AG presence	-0.300***	-0.182***	-0.475***
	(0.057)	(0.021)	(0.057)
Brand × AG presence	-0.483***	0.004	-0.497***
	(0.083)	(0.031)	(0.083)
Molecule-Formulation Fixed Effects	Yes	Yes	Yes
Obs.	30,822	30,822	30,822

Notes:

Finally we look at the generic entry rate in our dataset. We define the "generic entry rate" at any quarter since LOE as the number of generic entries happening in that period in our sample, divided by the total number of generic entries observed in our sample (1652). Figure 4 shows the generic entry rate by quarters since loss-of-exclusivity. The most notable pattern is that a large fraction of the generic entry happens in the quarter when LOE happens (around 33% of all generic entries). This suggests that a fraction of generic entrants correctly anticipate LOE and get approval before the LOE period. However, the majority of generic entry happens after the first quarter. By quarter 20 after LOE, about 93% of the generic entries have taken place (Figure 7).

One possible issue could have been that the generics all received final approval as soon

Figure 3: Generic entry rate.



(a) Fraction of total generics entering in every quarter since LOE.

(b) Cumulative fraction of total generics entering in every quarter since LOE.

as LOE occurs, and the variation in entry times is purely manufacturing frictions that is staggering out product entries over time. This can be ruled out for sevverals reaons. First, Wang et al. (2022) use exact dates on when a generic receives FDA approval, and identifies that date as "entry". From their paper, "In the data we observe that around 40% of the market entries occurred within the first year, and more than half of the entries happened within the first two years after the patent expiry." Therefore the majority of generic approvals in fact happen after the quarter of LOE. Second, generics generally start selling their products as soon as they receive approval, which makes sense as they would want to start recouping their sunk cost of entry and the earlier periods of the market are the most profitable.

4 Model

To study market dynamics after loss of exclusivity, we develop a structural model of entry and price competition between authorized generics and generics. The aim is to develop a stylized model that is tractable enough for estimation but captures the important economic mechanisms. On the demand side, we set up a discrete choice model of demand, where the consumer first chooses a group (brand, nonbrand, or outside option) then chooses a product within the group. On the supply side, we construct a two-stage model. The first stage is a static game of generic entry. The second stage is a dynamic game of Authorized Generic release decision. Every period, generics who decided to enter are stochastically approved for entry by the FDA, the branded drug manufacturer decides whether to release an AG, and the market participants compete in prices.

4.1 Demand

We model consumer demand as proceeding in two stages. In Stage 1, for a given molecule-formulation, each consumer chooses a group – brand, nonbrand, or outside option (forgoing medication entirely, switching to another molecule, etc.). The non-brand group includes both generics and AG (if present). In Stage 2, each consumer picks a product from the group chosen in Stage 1. Consumer choice is driven by the characteristics and prices of each product.

Stage 1: For molecule-formulation m, let g denote a group. Thus:

$$g \in \{\text{brand}, \text{non-brand}, \text{outside option}\}$$

In Stage 1, the Consuming Agent (a mix of the patient, provider, and insurer) chooses a group. The utility of consuming agent i for group g in market m at time t is given by:

$$u_{igmt} = \gamma_m^{(1)} + \lambda_t^{(1)} + \alpha_i^{(1)} ln\, p_{gt} + \beta_{1i}^{(1)} \text{brand}_g + \beta_2^{(1)} brand_g \cdot \text{time-since-loe}_t + \xi_{gt}^{(1)} + \varepsilon_{igt} \tag{1}$$

where $ln\, p_{gt}$ denotes the average of log prices within the group, "brand" is an indicator for the brand group, and $\xi_{gt}^{(1)}$ allows for unobserved group-specific quality. The variable "time-since-loe" measures the number of quarters since generic entry (loss of exclusivity), and this is used to form a brand-specific time trend.[13] The random coefficients $\alpha_i^{(1)}$ and $\beta_i^{(1)}$ allow for heterogeneity in price sensitivity and brand valuation respectively. As is common in the empirical literature, we further assume $\alpha_i^{(1)} \sim \mathcal{N}(\alpha^{(1)}, \sigma_\alpha^2)$ and $\beta_i^{(1)} \sim \mathcal{N}(\beta_1^{(1)}, \sigma_1^2)$.

Note that the brand group has a single product, while the non-brand group can have

[13]Branded drug manufacturers can sometimes take actions that slow down generic adoption. This involves things like product hopping, getting approval for new therapeutic uses, gaining orphan drug designation, etc. These moves stop consumers and pharmacists from switching over to generic drugs and get around automatic substitution laws. While our model does not specifically account for such actions, our demand specification contains a brand-specific time trend that aims to capture the idea that the branded drug loses its appeal over time as the manufacturer runs out of such tactics. As would be expected, we find that the branded drug becomes less attractive over time, possibly because such tactics become less effective over time.

multiple products. Thus $ln\, p_{gt}$ is the branded drug's price for the brand group, and is the *mean* of all non-branded drugs' prices for the non-brand group. The price of the outside option is normalized to zero as is standard in the discrete-choice estimation literature.

Stage 2: If the Consuming Agent chose the non-brand group in Stage 1, and there are multiple non-brand drugs available for the molecule formulation m at time t, then the patient has to choose from among these products.

In Stage 2, the Distributing Agent (a mix of the pharmacy, procurement officer at the provider, and other intermediaries) help the patient choose a specific non-brand product. The utility of Distributing Agent i choosing non-brand product j in market m at time t is given by:

$$u_{ijt} = \gamma^{(2)}_{m(j)} + \alpha^{(2)} ln\, p_{jt} + \beta^{(2)}_1 \mathrm{AG}_j + \xi^{(2)}_{jt} + \varepsilon_{ijt} \tag{2}$$

where $ln\, p_{jt}$ denotes the log price of product j, AG_j is a dummy for product j being an Authorized Generic, and $\xi^{(2)}_{jt}$ allows for unobserved product-specific quality.

The market share of group g is given by integrating over the distribution of random coefficients:

$$s_g = \int \frac{e^{\delta^{(1)}_g + \mu_{ig}}}{\sum_l e^{\delta^{(1)}_l + \mu_{il}}} f(\mu_i|\tilde{\theta}_2) d\mu_i \tag{3}$$

where $\theta_2 = (\sigma_\alpha, \sigma_\beta)$, $\delta^{(1)}_g = \gamma^{(1)}_m + \lambda^{(1)}_t + \alpha^{(1)} ln\, p_g + \beta^{(1)}_1 \mathrm{brand}_g + \beta^{(1)}_2 brand_g \cdot \text{time-since-loe} + \xi^{(1)}_g$, and $\mu_{ig} = \sigma_\alpha \nu_{\alpha,i} ln\, p_g + \sigma_\beta \nu_{\beta,i} \mathrm{brand}_g$.

Conditional on choosing nonbrand group, the probability of a consumer choosing product j from nonbrand group is given by:

$$s_{j|g} = \frac{e^{\delta^{(2)}_j}}{\sum_k e^{\delta^{(2)}_k}} \tag{4}$$

where $\delta^{(2)}_k = \gamma^{(2)}_{m(k)} + \beta^{(2)} \mathrm{AG}_k + \alpha^{(2)} ln(p_k) + \xi^{(2)}_k$.

Stage 1 gives the expression for the market share of a group, while Stage 2 gives the expression for the market share of a product conditional on a group. Therefore, we can express the market share of a single good in terms of group market share and market share of good conditional on choice of group:

$$s_j = s_{g(j)} s_{j|g(j)} \tag{5}$$

See the Appendix B for a detailed derivation and discussion of Equation 5. Therefore, estimating the parameters in Stage 1 allows us to predict $s_{g(j)}$, and estimating the parameters in Stage 2 allows us to predict $s_{j|g(j)}$. This allows us to study counterfactual scenarios and how they affect market shares of all products. Futhermore, estimating this demand system amounts to estimtaing a random-coefficient discrete choice model over groups, then a multinomial logit model over the nonbrand products.

4.1.1 Discussion on demand model

We now give a discussion of our modeling choice for the demand system, specifically why we choose the two-stage demand setup instead of other demand setups. Generally, papers estimating demand systems follow the setup of Berry et al. (1995), where each consumer chooses from a set of products. When trying to account for correlated preferences within a group, some papers use a random-coefficients or nested logit setup . In contrast, in our setting, a consumer chooses a group, then chooses a product within a group. The reasoning for this is that we found, after experimenting with different specifications, that this matched the empirical patterns of prices and market share best. In our data we see that the price difference between the brand and non-brand group leads to substantial switching from branded drug to non-branded drugs, which implies high price elasticity of consumers. However, within non-brand products, we see very little switching based on specific prices of non-brand drugs. That is, a higher-priced generic has only slightly lower market share than a lower-priced generic. This then suggests consumers are not price elastic. When all products are estimated together in a single discrete choice setting like Berry et al. (1995), we end up finding very low price-elasticity overall for the market. When we do counterfactuals, this low price-elasticity leads to unrealistic predictions such as low substitution between brand and generics after loss-of-exclusivity (which is at odds with what we see in the data).

Our specification gets around this by capturing the right substitution patterns in the data. Stage 1 captures the fact that the price gap between brand and nonbrand leads to a large substitution towards the cheaper nonbrand products. Stage 2 picks up the mild price elasticity between generics, while also helping explain why average prices decline as more generics enter the market. While our demand model is somewhat different from other discete-choice settings, note that our aim for the project is to study entry decisions; as long as we correctly predict prices and per-period profits, our central arguments work.[14]

4.2 Supply

We first present an overview of our supply model. The following subsections explain the components of the model, and Section 4.3 goes over the reasoning behind the modeling choices.

Recall that generic manufacturers have to wait for FDA approval before they launch for approval; as a result, they enter (apply for ANDA approval) well in advance. In contrast, authorized generics can be released at any time. This motivates our model setup below.

There are two stages to the supply side:

1. In the first stage, generic firms decide whether to apply to enter a market (i.e. apply for ANDA to molecule-formulation) or not. This stage is modeled as a game of simultaneous entry application between identical firms.[15]

2. In the second stage, loss of exclusivity happens and a dynamic game begins. The dynamic game lasts T periods. Every period, a stochastic number of the generic firms which applied for entry are approved and enter into the market. The branded drug manufacturer decides whether to release AG or not (where the AG release decision is irreversible). Then, all the firms set prices and receive payoffs.

We now explain the model in reverse, starting from payoffs in the second stage.

4.2.1 Per-period payoffs

The branded drug manufacturer's per-period payoff is:

[14] Another option is to use a flexible regression on observed revenues to predict firm revenues in different states. An example of such a regression is presented in Table 2.

[15] An alternative way of doing this would be to assume sequential entry, i.e. firms apply for ANDA sequentially until the value of entering no longer exceeds cost. Under our assumption of identical generics, both simultaneous and sequential entry application will yield the same number of generic entrants.

$$\pi^b(s_{mt}) = [P^b_{mt} - MC^b_m]s_b(s_{mt})M_m + \mathbf{1}(AG_{mt} = 1)\left[[P^{AG}_{mt} - MC^{AG}_m]s_{AG}(s_{mt})M_m\right] \quad (6)$$

where s_{mt} denotes the state variable vector in market m, superscripts and subscripts b and AG refer to the branded product and AG respectively, P denotes price, MC denotes marginal cost, $s(s_{mt})$ denotes market share, and M_m denotes market size. The indicator variable $\mathbf{1}(AG_{mt} = 1)$ denotes whether an AG has been released onto the market. The intuition behind the branded drug manufacturer's payoff is that it makes the first half of the payoff from the branded drug, and the second half from the AG conditional on releasing the AG.

Similarly, the generic firm l's per-period payoff with the firm-specific state variable vector $s_{l,t}$ is:

$$\pi^g(s_{lmt}) = (P^g_{mt} - MC^g_m)s_g(s_{lmt})M_m \quad (7)$$

The branded drug, generics and AG compete on prices. We use a reduced-form price prediction regression to predict prices in different states and markets; see Section 5.2. While the model has been written for maximum generality, during estimation we set $MC_b = MC_{AG} = MC_g$ for a given market m. This assumption can be relaxed.

4.2.2 Second-stage

A dynamic game begins from $t = 1$ and lasts T periods, where every period is a quarter. Let n^*_e be the number of generic firms that have applied for an ANDA (which is determined in the first stage). In period $t = 2$ the branded drug's market exclusivity expires, and every period a random number of generic firms gain FDA approval and enter the market.

Since AG release is a dynamic decision, the branded drug manufacturer's choice problem can be written in terms of value functions. Thus, the value function for a branded drug manufacturer every period is given by:

$$\begin{aligned} V^b(s_{mt}, \varepsilon_{mt}) = \max_{AG_{mt+1}\in\{0,1\}} \pi^b(s_{mt}) - \mathbf{1}(AG_{mt} = 0, AG_{mt+1} = 1)\kappa^{AG}_m + \\ \beta E[V^b(s_{mt+1}, \varepsilon_{mt+1})|s_{mt}, \varepsilon_{mt}] + \varepsilon_{mt}(AG_{mt+1}) \quad (8) \end{aligned}$$

where $AG_{mt+1} = 1$ means the AG is in the market m during period $t+1$. AG entry cost in molecule-formulation m is denoted as κ_m^{AG}. The AG entry decision is irreversible, i.e. $AG_{mt} = 1$ implies $AG_{mt+k} = 1\ \forall k > 0$. AG entry is implemented with a one-period delay.

The timing of AG release could be influenced by factors observed to the manufacturer but unobserved to the econometrician. To account for such factors, the model (following the existing literature on dynamic games) includes choice-specific payoff shocks $\varepsilon_{mt}(AG_{mt+1})$. These shocks follow a Type 1 Extreme Value distribution.

The probability of releasing an AG conditional on not having yet released it is given by[16]:

$$Pr(AG_{mt+1} = 1|s_{mt}, AG_{mt} = 0) = \exp\left(\frac{-\kappa_m^{AG} + E\left[V^b(s_{mt+1}, AG_{mt} = 1|s_{mt}, AG_{mt} = 0)\right]}{\sigma_\varepsilon}\right)/B \quad (9)$$

where

$$B = \exp\left(-\kappa_m^{AG} + E\left[V^b(s_{mt+1}, AG_{mt+1} = 1|s_{mt}, AG_{mt} = 0)\right]\right) + \exp\left(E\left[V^b(s_{mt+1}, AG_{mt+1} = 0|s_{mt}, AG_{mt} = 0)\right]\right)$$

AG release is an irreversible decision:

$$Pr(AG_{mt+1} = 1|s_{mt}, AG_{mt} = 1) = 1 \quad (10)$$

Similarly, the value function for generic l is given by:

$$V^g(s_{lmt}) = \pi^g(s_{lmt}) + \beta E[V^g(s_{lmt+1})|s_{lmt}] \quad (11)$$

where $s_{l,t}$ includes whether generic l has been approved for production by the FDA. While the generic manufacturers do not face a dynamic decision in the second stage, they use the value function to make entry decision in the first stage.

After period T, the market state is set at s_{mT}, and the manufacturer receives the corre-

[16]This expression involves a slight abuse of notation: s_{mt} already includes the value of AG_{mt}, but we separate them out in the arguments here for clarity.

sponding payoff for infinite periods:

$$V^b(s_{mT}) = \sum_{\tau=T}^{\infty} \beta^\tau \pi^b(s_{mT}) \tag{12}$$

Similarly for generics:

$$V^g(s_{mT}) = \sum_{\tau=T}^{\infty} \beta^\tau \pi^g(s_{mT}) \tag{13}$$

The manufacturers have to form expectations over generics' approval probabilities; recall that approval is given randomly to entrants over time. To express the value functions, we need to pin down how state transitions happen regarding generic approval. Thus, we model the FDA approval rate as a binomial process. This follows Ching (2010).

Suppose by period t in market m, $\mathcal{N}_{mt}$ generic entrants have already received approval and launched their products. Thus $n^*_{em} - \mathcal{N}_{mt}$ entrants have not yet received approval . The probability that of the $n^*_{em} - \mathcal{N}_{mt}$ remaining entrants, k will receive approval in this period is given by:

$$P_e(k, \mathcal{N}_{mt}, t) = \binom{n^*_{em} - \mathcal{N}_{mt}}{k} \lambda(t)^k (1 - \lambda(t))^{n^*_{em} - \mathcal{N}_{mt} - k} \tag{14}$$

where $\lambda(t)$ – the probability of a single generic receiving approval – is estimated from the data. We assume the branded drug manufacturer knows n^*_e and total generics that received approval for $t = 2$ ($\mathcal{N}_{m2}$).

4.2.3 First-stage

In the first stage, an infinite number of generics decide whether they want to enter. We assume all generic firms are ex-ante identical, do not receive private error draws for entering and staying out, and do not know their draws of ξ_{jt} conditional on entry.

Generics keep entering until the value from entering no longer exceeds the cost of entry, i.e. free entry condition as per Igami (2018). More precisely, the equilibrium number of generic entrants is n^*_e, determined by the following inequality:

$$V^g(s_{gm0}, n^*_{em}) \geq \kappa^g_m > V^g(s_{gm0}, n^*_{em} + 1) \tag{15}$$

where κ^g_m denotes the generic manufacturer's entry cost for molecule-formulation m and

s_{gm0} is the initial state of the market from a generic's perspective.. Intuitively, the value function of entry is declining in the number of entrants - more entrants mean more competition in the market and lower per-period profits. At the equilibrium n_e^*, current entrants can cover their entry cost, but an additional entrant would stop that from happening. This partly follows Igami (2018). Finally, unlike the branded drug manufacturer, the generics don't know $\mathcal{N}_{m2}$.

4.2.4 Model solution

We solve the model by backward induction and recover the equilibrium number of generic entrants and conditional choice probabilities (CCPs) of authorized generics. Since the AG's CCPs enter the value functions of generic entrants and the AG knows n^*, all agents are forming rational expectations about each others' choices when making a decision. This means that a policy or environment change that affects the choice probabilities of any agent will have feedback effect on other agents in the market. The details of the model solution are given in Appendix A.1.

4.3 Discussion

Economic interpretation of AG choice-specific shocks and their variance: Recall that the AG release decision faces choice-specific payoff shocks $\varepsilon_t(AG_{t+1})$ that follow a Type 1 Extreme Value distribution. These logit shocks have a logit scaling parameter σ_ε that governs the variance of these shocks. Intuitively, this is used to account for payoff shocks observed to the agent but unobserved to the econometrician.[17] These payoff-shocks could subsume a variety of factors such as unobserved constraints, manufacturing delays, cost issues, organizational an transactional frictions, etc.[18] In our setting, it also subsumes cases

[17]During estimation, it is used to reconcile why we observe firms in similar states of the world make different choices. This is a common methodology used in dynamic discrete choice models with dollar-valued payoffs. See Igami (2018) and Alam (2023). Papers which have unitless payoff functions instead normalize the logit scaling parameter to 1.0 and estimate all other structural parameters relative to it, as in Rust (1987).

[18]A more detailed list of such factors is as follows. First, this could be driven by manufacturing delays, organizational frictions, and contracting frictions (if AG production is being outsourced). It may be that the loss-of-exclusivity was unanticipated or there was uncertainty surrounding it, which meant that the branded manufacturer was unprepared for the loss-of-exclusivity when it happens. There could be policy uncertainty such as how the FDA would approve or regulate the new generic entrants, or how a lawsuit might be resolved, that may add further uncertainty to the release decision. It could be that the branded manufacturer is engaging

where an AG was not released due to a No-authorized-generic settlement.[19] This paper does not model No-AG settlements explicitly, mainly because it is extremely difficult to identify when such settlements have happened. Our demand model is not affected by No-AG settlements, and our counterfactuals apply to markets where such settlements have not occurred. We include a brief discussion on No-AG settlements through the lens of our model in Section 7.4. In addition, see FTC (2011) for a more detailed analysis. Finally, the logit shocks subsumes the brand manufacturer engaging in other post-LOE strategies (such as product line extension or product discontinuations) that affect its AG release decision.

Discussion on price prediction equation: Instead of assuming a pricing conduct such as Nash-Bertrand or Nash-in-Nash bargaining, we use a price prediction regression to predict prices charged by different firms in different markets. We now discuss our reasoning for this approach and the trade-offs involved.

One pricing conduct assumption we could have made is Nash-Bertrand pricing, which however implies that all bargaining power is upstream with the pharmaceutical manufacturer. This is unlikely to be true in reality, since there are many intermediaries - Pharmacy Benefit Managers (PBMs), wholesalers, pharmacies, providers, etc. - that negotiate drug prices with substantial bargaining leverage. This is also borne up by the estimated price elasticities in our paper - we find most price elasticities to be less than 1, which aligns with a model of split bargaining power between upstream and downstream firms rather than a take-it-or-leave-it offer (TIOLI) from upstream firms. Since we have aggregate data instead of detailed plan-level or PBM-level data, we cannot explicitly model the pricing game between manufacturers and all the intermediaries.[20] The advantage is that our pre-

in other generic-deterring strategies (such as product-line extensions) which affects its AG timing decision. The branded drug manufacturer may not have any generic subsidiaries, which makes it very difficult to release an AG, because marketing a non-branded drug versus a branded drug entail very different skillsets. The marketing of a branded drug focuses on advertising and detailing, negotiating formulary positions by insurer, etc. The marketing of a generic drug focuses on extreme cost efficiency, winning pharmacy auctions (Cuddy (2020)), etc. These are very different skillsets, and for a branded drug, it may not be worth the hassle of getting into the generics business just to release an authorized generic. It could instead outsource its production to another generic firm (for instance, Prasco and Greenstone specialize in bringing authorized generics for other firms into the market) but that involves negotiation cost and profit-sharing.

[19] "No-AG settlements" are settlements where the generic firm agrees to not enter immediately in exchange for assurance that an Authorized Generic will not be released. Such a No-authorized-generic-settlement could be related to a Paragraph IV settlement regarding the same molecule-formulation, or a settlement involving a different molecule-formulation but involving the same firms.

[20] Other papers which study pharmaceuticals with aggregate data and make a pricing conduct assumption

dicted per-period profits are not susceptible to misspecifications about the pricing conduct assumption.[21]

Note that in our counterfactuals, we do not change any primitives that would affect the static pricing game. Otherwise, the price prediction equation estimated from the observed data would no longer be a good predictor, as a change in demand primitives should change the underlying pricing patterns. Instead, we limit our counterfactuals to changing parameters that affect dynamic incentives (such as entry restrictions and faster approval rates).

Our price prediction equation is also similar in spirit to the price-setting equation in Maini and Pammolli (2023). In their paper, the authors try to predict prices across different European countries with different bargaining policies and unknown objective functions. Instead of taking a strong stand on either, the authors opt for an agnostic approach to predicting prices and use a flexible control function. Our motivation is similar in that we want to be agnostic about how different intermediaries interact across the supply chain. In both our paper and Maini and Pammolli (2023), the main focus is on entry instead of the static pricing game, which means being able to accurately predict per-period payoffs is sufficient for our analysis.

Another issue is that during this period, some generic manufacturers may have been engaging in price-fixing (Cuddy (2020), Starc and Wollmann (2023)). This occurred from

include Arcidiacono et al. (2013), Dubois et al. (2022), and Starc and Wollmann (2023). Unlike these papers there are major differences in our analysis that preclude us from doing the same. Arcidiacono et al. (2013) focus on a small number of molecules, which allows them to leverage additional data and impute rebates. In contrast, our paper considers well over 200 molecule-formulations. Dubois et al. (2022) restrict their attention to sales in the hospital channel for drug sales, while we consider all channels. They assume Nash-Bertrand, which is a more reasonable assumption in their setting because hospitals are not covered by large intermediaries such as PBMs, and price elasticities exhibited in these non-patient-facing channels are much larger. Starc and Wollmann (2023) consider the pharmacy channel and use different datasets compared to this paper.

[21] A regression of prices on level of competition may suffer from endogeneity issues from demand shocks. This is less of an issue than in other settings, because generics cannot simply respond to a demand shock and enter the market - they need to wait years to get FDA approval for marketing. Furthermore, we also add in molecule and formulation fixed effects.

We also estimate marginal cost to match entry patterns instead of inverting a first-order condition involving observed prices, as is done in (among others) Dubois et al. (2022) and Starc and Wollmann (2023). If the pricing conduct assumption is incorrect, the imputed marginal costs would be incorrect as well, which we are avoiding. Other approaches by authors who do not impute marginal cost by assuming a pricing conduct such as Nash-Bertrand include Tunçel (2024) (who uses the minimum observed price to construct moment inequalities about non-negative costs) and Grennan et al. (2021) (who set marginal cost as 0.17 times generic price).

late 2013 onwards; our dataset's coverage is from 2004-2016. As a result, we rerun the price prediction regression by dropping years 2013 onwards, and obtain very similar results.

Generic entry as a static entry game: Recall that we model the entry decision by generics to be a static entry game that happens before loss-of-exclusivity (following Ching (2010)). We choose to model generic entry as a static game instead of a dynamic timing game for the following reasons. In the pharmaceuticals industry during our sample period of 2004-2016, the mean approval time for ANDA was about 40 months, so any decision to enter was implemented with significant and random delay. Note that the AG does not need FDA approval as it is riding on the branded drug's original approval. As a result, we model AG entry as a dynamic decision being made every period.

Another simplification we make is that while generics apply well in advance of patent expiration, they do not know exactly when they will get FDA approval. In reality there are sometimes cases where a generic firm applies for and gains ANDA approval well in advance of patent expiration, and so can start production in the same quarter when patent expires. Our estimation of generic entry rates partly accounts for that; see Section 5.3 for a detailed explanation.

Another concern could be that some generics file for entry well after loss-of-exclusivity. It turns out not to be much of an issue for our setting. Starc and Wollmann (2023) find that the mean ANDA filing in mature generic markets is 0.21 per drug-year (even lower for non-cartelized markets), and conclude that "entry into (mature) generic markets is typically rare".

Information assumptions: To simplify computation, we make two assumptions on information regarding the number of generic entrants. First, we assume that the AG knows how many generics have filed for ANDA to its molecule-formulation.[22] Second, we assume that the authorized generic correctly predicts the number of generic launches in the quarter when loss-of-exclusivity happens; this is realistic as the FDA would have announced before loss-of-exclusivity which ANDAs have been approved by then.

[22] In reality, the information on ANDA application by a generic manufacturer becomes public only after it wins ANDA approval from the FDA (although generic manufacuterers may have private information sources that could alert them to ANDA filings by rivals). This is a simplifying assumption that improves the tractability of our model.

5 Estimation

We use a reduced-form price regression to predict prices, and estimate generic approval rates from the data. Demand is estimated using methods from Berry et al. (1995) and Maggio et al. (2022). The supply model is solved via backward induction, and AG's logit scaling parameter is estimated via Maximum Likelihood.

5.1 Heterogeneity by channel presence

Our dataset contains a breakdown of drug sales by the channels were the sale happened. These channels are clinics, drugstores, federal facilities, food-stores, long term care hospital, HMO, home health care, mail services, and non-federal facilities (Berndt et al. (2017), Dubois and Majewska (2022)). In our analysis we want to be cognizant of the fact that price sensitivity and pricing behavior can vary across different distribution channels. More generally, there could be variation across channels based on how much input the patient has in choosing between brand versus non-branded drug.

To that end, we classify all channels into one of two types: patient-facing or non-patient-facing. "Patient-facing channels" are channels where the individual patient has more influence over group choice, whereas "non-patient facing channels" are where patients have limited influence. For instance, a patient buying from a pharmacy is allowed to specifically purchase a branded drug, subject to additional requirements. However, a patient being administered medication at a hospital has much less influence over group choice. We can expect demand parameters to vary between these two categories because of the intermediaries and bargaining processes involved. In patient-facing channels, the insurer, patient, physician, and PBM together determine the product choice and prices; by contrast, non-patient-facing channels involve the provider negotiating with the wholesaler or manufacturer and determining product choice. The channels we classify as "patient-facing" are retail pharmacies, mail pharmacies, and food stores; the remaining channels are classified as "non-patient facing".

A molecule-formulation is classified as patient-facing if more than 50% of its sales are in patient-facing channels. In our data, we classify 183 molecule-formulations as patient-facing and 58 molecule-formulations as non-patient-facing.

5.2 Price prediction and generic approval rates

Instead of assuming a pricing conduct (such as Nash-Bertrand or Nash-in-Nash bargaining), we choose to be agnostic about the price-setting behavior and instead predict prices using observables. More specifically, we regress observed prices on covariates such as molecule fixed effects, formulation fixed effects, number of generics, presence of branded drugs, etc. This regression is then used to predict prices in different states of the world. We use the regression reported in Table 3 for this purpose. See Section 4.3 for a detailed discussion on why we do this instead of making a pricing conduct assumption such as Nash-Bertrand.

We make several simplifying assumptions. First, we assume that all non-brand drugs charge the same price in a market-quarter[23] Second, we leave out quarter fixed effects in our demand system, and instead add the median estimated quarter fixed effect to the demand system's intercept term.

We combine this price prediction equation and estimated marginal costs with our estimated market share equation (Equation (5)) to generate per-period profits in different states.

5.3 Estimating generic entry rates

To operationalize Equation (14), we estimate the average generic entry rates $\lambda(t)$ at every quarter since loss-of-exclusivity (LOE). This is done as follows. For each molecule-formulation and quarter since LOE, we calculate the number of generics waiting to enter, i.e. final number of generics observed in that molecule-formulation minus the current number of generics present in the molecule-formulation. We sum this number across all molecule-formulations at a given quarter since LOE to get the remaining entrants for every quarter since loss-of-exclusivity in our dataset. Then we find the total generic entry that happens in every each molecule-formulation and quarter since loss-of-exclusivity.[24] We sum this number across all molecule-formulations at a given quarter since LOE to get the total generic entry for every quarter since loss-of-exclusivity in our dataset. Taking the ratio

[23]In our data we see AGs price close to other generic manufacturers in the market, a fact also corroborated in FTC (2011).

[24]A generic manufacturer's entry into a molecule-formulation is identified as the earliest quarter that manufacturer is spotted in our dataset for that molecule-formulation.

Table 3: Price prediction regressions

	Patient-facing	Non-patient-facing
formulation: Injectable	2.118	–
	(0.136)	(–)
formulation: Oral	-1.308	–
	(0.112)	(–)
No. of nonbrand drugs	-0.269	-0.298
	(0.009)	(0.016)
No. of nonbrand drugs squared	0.008	0.010
	(4.843e-04)	(0.001)
Brand	0.123	-0.076
	(0.031)	(0.041)
Brand * time-since-generic-entry	0.017	-0.005
	(0.001)	(0.002)
Brand * No. of nonbrand drugs	0.232	0.231
	(0.004)	(0.008)
Brand * Authorized generic present	0.041	0.774
	(0.031)	(0.064)

Notes: We regress log of prices on the covariates listed above. Robust standard errors are in parentheses.

of total generic entry by remaining entrants at every quarter since loss-of-exclusivity gives us the average generic launch rates at every quarter since loss-of-exclusivity.[25]

Note that the estimated generic launch rates comprise not just the FDA's approval rate but also generics applying early in an effort to get FDA approval close to the loss-of-exclusivity. More specifically, a number of generics apply for and receive FDA approval well in advance of loss-of-exclusivity, and then wait for the loss-of-exclusivity to happen before entering the market. This is why we see over 30% of the generic entries happening in the very first quarter. In general, the distribution of generic entry rates does contain information about the timing of the generic application itself, which we do not observe and therefore do not model.

This explains why we find generic launch rates to be highest at the beginning of loss-of-exclusivity, and declining over time; it reflects the idea of anticipating loss-of-exclusivity described above. Once enough time has passed, the generic entry rate rises up to 1.0 because all applications are eventually resolved. We implicitly rule out cases where a generic applies for entry and never receives approval.

So why is this measure of generic launch rates correct for our setting? First, this is the correct empirical probability distribution that generics use to form beliefs about their entry timing. Second, regulatory frictions are subsumed in this measure. Therefore, if regulatory frictions were reduced, that can be interpreted as the generic entry rates at each period being higher.[26]

[25] In our calculation of generic entry rates, we limit the number of entrants to those happening on or before 17 quarters. Our main worry is that entry happening after 17 quarters could involve generics who applied well after loss-of-exclusivity; filings happening after LOE could be for other reasons not modeled in our framework. Starc and Wollmann (2023) find that the majority of ANDA filings are approved within 4 years of filing, hence we limit our analysis to 17 quarters, and robustness checks with 24 quarters are very similar. It turns out to matter little for our results because there are few generic entries happening after 17/24 quarters in any case.

[26] Note that the majority of generic entries happen after the first period. If FDA approval rates were near-immediate, we would expect the majority of generic entries to happen in the first period. It is well-established that the earliest periods of the market are the most profitable for generic entrants. For instance, this is the reason behind the 180-day exclusivity period of the Hatch-Waxman Act to incentivize Paragraph IV certifications. It also explains why generics are strongly against AG launches (particularly during 180-day exclusivity periods) as it introduces competition in the most profitable early periods of the market.

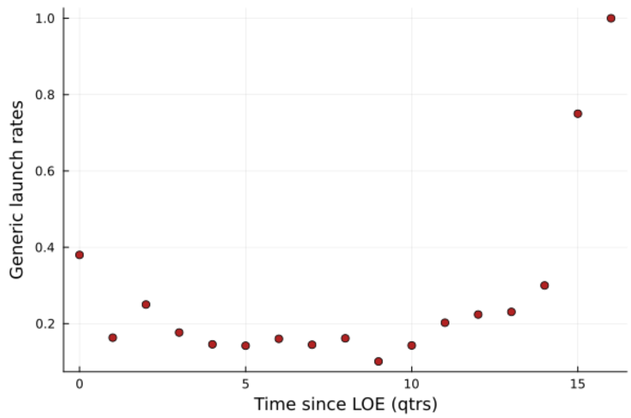


Figure 4: Generic entry rate

5.4 Demand estimation

We estimate each stage of the demand model separately. Stage 1 of the demand model is a standard random-coefficients discrete choice model, so we use the method of Berry et al. (1995) (see Conlon and Gortmaker (2020) and Nevo (2000) for detailed discussions on implementation). We use Gandhi-Houde instrumental variables (Gandhi and Houde (2019)) and employ two-step GMM. We allow our demand parameters to vary based on whether the molecule-formulation is "patient-facing" or "non-patient facing".

Stage 2 of the demand model is a multinomial logit demand model but without an outside option. Given that idiosyncratic errors follow Type-1 Extreme Value distribution, we can write market share of generic drug j conditional on nonbrand group being chosen in Stage 1, as:

$$s_{jt|g} = \frac{e^{\delta_{jt}^{(2)}}}{\sum_k e^{\delta_{kt}^{(2)}}}$$

where $\delta_{kt}^{(2)} = \gamma_{m(k)}^{(2)} + \beta^{(2)}\mathrm{AG}_k + \alpha^{(2)} ln(p_{kt}) + \xi_{kt}^{(2)}$. Using the transformation from Berry (1994):

$$ln(s_{jt|g}) = -ln(\sum_k e^{\delta_{kt}^{(2)}}) + \delta_{jt}^{(2)}$$

Plugging in the formula for $\delta_{jt}^{(2)}$ gives us the estimating equation:

$$ln(s_{jt|g}) = -ln(\sum_k e^{\delta_{kt}^{(2)}}) + \gamma_{m(j)}^{(2)} + \beta^{(2)}\mathrm{AG}_j + \alpha_2 ln(p_{jt}) + \xi_{jt}^{(2)} \tag{16}$$

Following Maggio et al. (2022) this can be estimated using a linear regression of logs of market share on market-by-quarter fixed effects and other covariates. That is, the first term on the right hand side varies at the market-by-quarter level, so including market-by-quarter fixed effects absorbs it. The remaining coefficients can just be estimated through a linear regression with fixed effects ($\xi_{jt}^{(2)}$ becomes the econometric error in the regression).[27]

5.5 Supply estimation

We estimate the supply-side cost parameters of our model. First we estimate marginal costs by molecule-formulation by matching predicted generic entry numbers to observed numbers at publicly reported generic entry costs. Next, we estimate the logit scaling parameter of the AG choice-specific shocks. Throughout we set the quarterly discount factor at $\beta = 0.95$.[28]

First we derive approximate bounds on the marginal costs by matching predicted generic entry numbers to observed numbers at publicly reported generic entry costs for markets with no AGs. Recall that we do not make assumptions about pricing conduct such as Nash-Bertrand to avoid misspecification; as a result, we cannot impute marginal costs from observed prices (though as we will see later, we do use the average observed prices in the molecule-formulation to inform the marginal costs of that molecule-formulation). Instead, our aim is to instead estimate marginal costs from observed generic entry numbers.

[27] By contrast, in Berry (1994) the authors use the outside good's market share to net out the term $ln(\sum_k e^{\delta_{kt}})$. In our case this cannot be done because there is no outside option in Stage 2; conditional on choosing the non-brand group in Stage 1, the consumer has to choose from one of the non-brand products in Stage 2.

[28] This lines up with an annual discount factor of 0.81, which has been used in prior empirical work such as Igami and Sugaya (2022). Some dynamics papers use higher annual discount factor such as 0.95. We choose a lower number because i) we do not account for exit or discontinuation of manufacturers, so the low discount factor may indirectly account for that possibility by downweighting future payoffs more, ii) pharmaceutical industry sees new molecule-formulations experiencing loss-of-exclusivity regularly so there's a high risk of a product market becoming obsolete in the future. Such risks of product obsolescence is implicitly captured by a lower discount factor.

The intuition is that the estimated marginal costs will scale down predicted profits so that at the publicly-reported generic entry costs, the model predicts reasonable generic entry numbers.[29]

The estimation is done as follows. We assume that the marginal cost for molecule-formulation m can be approximated by $MC_m = \vartheta \bar{p}_{gm}$, where $\bar{p}_{gm}$ is the average observed price of generics in molecule-formulation m over all periods in our dataset, and ϑ is a scaling parameter. The gain from scaling the average price of generics to estimate marginal costs is that i) generics price close to the marginal cost, so there is information about marginal cost levels from observing their prices[30], and ii) it allows marginal cost to vary across molecule-formulations. The alternative of having the same marginal cost across all molecule-formulations would be too restrictive, and trying to estimate marginal cost by molecule-formulation is likely to be noisy and computationally costly.

We isolate molecule-formulations for which AGs do not enter, and assume that AGs had zero probability of entering these markets. For a given generic entry cost κ_g, we try different trial values of ϑ and solve for the equilibrium generic entrants in each market. Using grid-search, we pick the ϑ that minimizes the sum of squared differences between the observed and predicted generic entry numbers. Recall that reported generic entry costs vary between \$3 million and \$15 million; therefore we set κ_g to each of these bounds and recalculate ϑ. We find that ϑ range from 0.54 to around 0.65. Thus, we fix per-period fixed costs to be the average of these two bounds for our main estimation. Details of the estimation procedure are given in Appendix A.3.

We also assume for our AG cost estimation and counterfactuals that all manufacturers in a molecule-formulation have the same marginal cost, i.e. $MC_m^b = MC_m^g = MC_m^{AG} = MC_m$.

Next we estimate the logit scaling parameter of the authorized generics' choice-specific shocks σ_ε and set κ_m^{AG} to zero. We set $T = 32$ (meaning that the dynamic game lasts 32 quarters) and impose that profits become zero afterwards for reasons such as product obsolescence (both assumptions can be relaxed). The parameter is estimated via Maximum

[29]This suggests that our interpretation of marginal costs is somewhat broader than the cost of producing an additional pill. Our estimates of marginal costs of operation reflects the cost of operating the factory and production line, maintaining supply chains, transaction costs, etc. It is also the economic cost of marketing a product, and so reflects costs beyond accounting costs. For instance, there could be opportunity cost tied to allocating inputs to this product as opposed to elsewhere, or there could be additional constraints facing the managers.

[30]For example, Grennan et al. (2021) set marginal cost as 0.17 times generic price.

Likelihood Estimation.

This raises the question of why we set κ_m^{AG} to zero. First, the generic's sunk cost of entry mostly comes from having to reverse-engineer the production process of the branded drug and setting up the supply chains; this is not a cost the AG release has to incur. Other costs such as marketing, production, and distribution are captured in the estimated marginal costs. Second, we conduct robustness checks by setting κ_m^{AG} to vary between 0 to \$3 million (which is the lower bound of generic entry costs) and find the estimated σ_ε to be very similar across all these values.[31]

6 Results

Tables 4 and 5 show the results of the demand estimation. From Table 4 we see that price has a negative impact on demand, and that the price-sensitivity is stronger for non-patient-facing molecule-formulations than patient-facing molecule-formulations. We also estimate the random coefficients to have statistically insignificant variance, meaning that any heterogeneity on the demand side is sufficiently picked up by the logit demand shocks. Results of estimation of the logit scaling parameter of AGs is reported in Table 6.

7 Counterfactuals

In this section we first explain how the counterfactuals are conducted then discuss the results of our counterfactuals.

To understand our model and predict the impact of different policies, we solve the model by backward induction. For the counterfactuals, we solve for both the number of generic entrants and CCPs of AG. A description of the full-solution method is given in Appendix A.1.

Next we explain how we conduct and report our counterfactuals. For each counterfactual we solve for the model then run 200 simulations, then report the average results from the simulations. We run the counterfactuals for 200 of the 241 market-specifications (molecule-formulations) found in our data, and study the distribution of effects. Note that

[31] We also attempted to estimate κ_m^{AG} and σ_ε jointly, but found that they were poorly identified in practice.

Table 4: Results of demand estimation for Stage 1.

	Patient-facing	Non-patient-facing
ln(price)	-1.004	-2.781
	(0.033)	(0.244)
Brand	0.799	1.033
	(0.069)	(0.250)
Brand * time-since-generic-entry	-0.095	-0.008
	(0.007)	(0.014)
RC std: Brand	-5.946e-07	-1.812e-07
	(6.120)	(6.598)
RC std: Price	3.434e-07	-2.025e-07
	(4.778)	(11.838)

Notes: The demand model is estimated using 2-step nonlinear GMM. Robust standard errors are reported in parentheses. "RC std: Brand" and "RC std: Price" refer to σ_1 and σ_α respectively. Market and quarter fixed effects are suppressed.

Table 5: Results of demand estimation for Stage 2.

	Patient-facing	Non-patient-facing
Authorized Generic	0.857	0.067
	(0.077)	(0.186)
ln(price)	-0.647	-1.286
	(0.023)	(0.068)

Notes: The demand model is estimated using a fixed-effects regression following Maggio et al. (2022). Robust standard errors are reported in parentheses.

Table 6: Results of cost estimation.

σ_ε	586497.139
	(17277.888)

Notes: Estimation done using Maximum Likelihood. Standard errors are reported in parentheses.

we think of counterfactuals under different market specifications as mainly checking how our model predictions vary for different realistic parameter values. We vary generic entry costs across market-specifications to ensure that the predicted number of generics is not below 1 and not greater than the maximum number observed in the data; more details of the implementation over market-specifications can be found in Appendix C.

We now present a brief discussion on profitability in different phases of a market's lifecycle. The early quarters of a molecule-formulation after loss-of-exclusivity are the most profitable for firms. During these periods, there are few firms in the market, resulting in high markups and high economic profits. As time goes by, more generics get FDA approval and enter the market, resulting in markups being driven close to zero and low economic profits. This provides another reason why generics want to launch earlier – the earlier phase of the market yield the highest profits and so contribute most towards covering the entry cost.

7.1 Deterrence through early-move versus early-launch

In this section, we illustrate how both the independent generics and authorized generic can deter each other's entry, but in different ways. Generics can deter AG release because they make entry decisions earlier; however, the AG can deter generic entry because it is able to launch earlier and provide price competition from the early stages of the market.

An independent generic has an early-move advantage. This is because a generic makes its entry decision and incurs its fixed cost in the first stage, before the authorized generic. This ability to move earlier means that the generic can deter an authorized generic from entering. Note that the authorized generic having a substantially lower entry cost than independent generics (conditional on drawing a trade shock) means that this entry deterrence effect is light.

On the other hand, the AG can deter the independent generic's entry through the fact that it can launch earlier. Since the authorized generic can launch its product early – during the more profitable phase of the market – it has a strong entry deterrence effect on the independent generic.

To showcase these economic forces, we perform the following two counterfactuals.

First, we show that the generics are able to deter AG release when AGs do not enjoy the launch-timing advantage over generics. We fix the following: i) Generics can launch immediately upon loss-of-exclusivity (at $t = 2$), so that the AG no longer enjoys a launch timing advantage over generics; ii) AG has a low logit scaling parameter so that the AG can launch immediately; iii) AGs have the same entry cost as generics, so that the AG no longer has a cost advantage; iv) AG and generics face the same demand primitives, so that neither enjoy any demand-side advantage over the other. Thus, in this counterfactual, the AG no longer has the early-launch advantage or the cost advantage. We find that across nearly all market-specifications, the AG is not launched. The results are summarized in Table 8.

Next, we show that AG is able to deter generics better when it has a stronger early-launch advantage. To do so, we vary the AG's early-launch advantage to see how it affects generic entry. First we run simulations where the AG has low logit scaling parameter, which makes it more likely that an AG can launch very early without any delays. Next we run simulations where the AG has a high logit scaling parameter, which makes it likely that the AG launches much later due to delays. We limit to market-specifications where the fraction of simulations seeing AG releases is greater than 90% and the mean AG release time decreases as the logit scaling parameter increases. We find that for over 30% of market-specifications, the AG can deter more generics when it can launch earlier; the remaining market-specifications see no change in number of generics. The results are summarized in Table 9.

7.2 Generic approval rates

We examine the effects of reducing regulatory frictions affecting generic entry. In our counterfactuals, we scale up the probability of ANDA being approved in every quarter after loss of exclusivity, and simulate the impact on market outcomes.

We find that faster generic approval rates leads to fewer generics entering, prices being lower in initial periods, prices being weakly higher in later periods, and that the effect is more muted if AGs may be present in the market.

Under faster FDA approval rates, there are opposing forces on a generic's entry decision. On the one hand, faster approval rates means the generic can market their drug earlier, and so has a longer time (till the terminal period) to make profits and cover their entry cost. This force would incentivize more generics to enter. On the other hand, faster approval rate means that when a generic does enter the market, more of its competitors are already in the market, so it will face greater competition and smaller profits upon entry. This force would disincentivize more generics from entering.

So which of these forces dominate? This is an open question we can study by simulating the market under faster ANDA approval rates.

For the first group of counterfactuals we leave AGs out of the market for clarity. For most market specifications we see that as generic approval rates increase, there are fewer generic entrants; for the remainder, there is no change in total generics. This means that the entry disincentive from earlier competition generally dominates the entry incentive from longer operation time in the industry.[32] The results are summarized in Table 10. An example for a single molecule-formulation is given in Table 7.

Our counterfactuals help illustrate why the second force dominates. Take a molecule-formulation where, if generics can enter immediately, then there will be 3 generic entrants. If an additional generic where to enter, the prices will be driven down so low that the additional generic will not be able to cover its entry cost. Now let's take the same market, but with a slower approval rate. The first 3 generics will still choose to enter, but their entries are now more spaced out over time. What about an additional fourth entrant? If the fourth entrant enters, then in the long-term the prices will be very low as in the immediate case. But in the short term, the entrant might get a lucky early approval and enter the market in its early periods when there's little competition, and make large profits. Thus, for the fourth entrant, the prospects of entering the market have actually improved with the slower approval rate. Slowing down the approval rate creates an element of luck where an additional entrant can make large profits in the early periods of the market.

Next we rerun the counterfactuals but allow the authorized generic to be released. Similar to before, for many market specifications, as ANDA approval rates increase, there are

[32] To see why, we can look at the per-period prices (and profits) in the industry. The earliest periods are the most profitable, because there are few nonbrand products in the market, resulting in high markups. As more generics enter, these markups are driven down to zero, so that in the long run price is nearly equal to marginal cost. This means that after a certain number of periods, generic firms no longer make economic profit that can go towards covering their entry cost. With faster ANDA approval rates, price drops to marginal cost much faster, meaning that there are far fewer quarters when generics can make economic profit, because there will be more competitors in the market right from the beginning. Thus, while it is true that generics can operate in the market longer due to being approved earlier, most of that time is going to be spent making zero economic profit.

fewer generic entrants. However, with AG, the decline in generics as approval rate increases is smaller; that is, as approval rate increases, the number of generics declines by a smaller amount with AG presence compared to without AGs. In fact, there are more market-specifications for which there is no change in number of generics. The results are summarized in Table 11.

While faster approval rates lead to fewer generics, it seems to have surprising effects on prices – prices are lower in the early periods of the market but are often higher in the later periods. Prices are lower in the early periods because, even though there are fewer generic entrants, most of them get released very quickly into the market after LOE. However, the fact that there are fewer generic entrants means that prices are often higher in the later periods of the market.

Table 7: Market outcomes with different generic approval rates for one molecule-formulation. The AG is prevented from entering the market for expositional clarity.

Cases	**Total generics**	**Mean generic price (early periods)**	**Mean generic price (later periods)**
Estimated	7.0	93.39	44.91
Estimated x 2	7.0	54.7	44.91
Immediate entry	6.0	53.28	53.28

Notes: Simulation conducted for market-specification of molecule-formulation Acamprosate-oral. “Estimated x *n*” refers to the estimated generic approval rates multiplied by *n*. “Immediate entry” refers to all generics entering the market immediately after loss of exclusivity.

7.3 Ban on Authorized Generic

We examine if AGs and similar fighting-brand strategies should be encouraged or banned by policymakers. To do so, we simulate market specifications under allowing AG to be released versus AG ban. We find that a ban on AGs leads to weakly more generic entrants – and sometimes multiple additional new generics – but has opposing effects on overall prices. We limit our counterfactuals to markets where over 90% of the simulations see an AG released. This is because it is only for these markets that the AG ban has an impact; in other market-specificiations, if an AG is unlikely to enter, then imposing the ban makes little impact.

In our counterfactuals we find that, when AGs are banned, nearly all market specifications see an increase in the number of generics. In particular, over 60% of the markets see multiple additional new generics entering. For the large majority of market-specifications, AG ban leads to higher prices in the early periods of the market. If the AG was deterring multiple generics from entering, prices are lower in the later periods of the market with an AG ban. The exact effect on number of generics and prices depends on the specific demand and supply parameters. The results are summarized in Table 12.[33]

We first explain the impact from an AG ban on generic entrants. There are two opposing forces operating on different phases of the market's timeline after loss-of-exclusivity:

1. Early AG launch: Without an AG ban, the AG is released early in the market, so there is always an additional competitor in the market from the early quarters. Without the AG, the generics face lower competition in the initial periods as all the generics slowly gain approval. Therefore, under an AG ban, prices are generally higher in the early phase of the market.

2. Greater generic entry: An AG ban leads to greater number of generic entrants. More generic manufacturers in the market drive prices down in the later phase of market.

The effect on later-period prices depends (among other things) on how many generics are deterred by the presence of an authorized generic. If an authorized generic ban leads to multiple new generics entering, then the drop in price in the later phase of the market more than offsets the missing price competition from the AG . As a result, in these markets the AG ban leads to lower prices in later periods. However if an AG ban leads to one or no new additional generic entrants, then AG ban leads to similar or higher prices in the later phase of the market .

Why can an AG ban lead to more than one new generic entering? This is because the AG is a more potent competitor than independent generics as a result of being released earlier in the market. Since the AG is released early, it has a negative impact on every generic and particularly during the most profitable early periods of the market. If the AG is banned, not only is there one fewer competitor, but it is also one less competitor in the most

[33] We also simulate the impact of banning AG entry only for the first few periods (1-2 quarters). We find that the ban leads to smaller generic deterrence but qualitatively same impact on prices. The results are summarized in Table 13.

profitable periods in the market. This results in a significant increase in expected generic profits compared to one fewer independent generic. In turn, this can incentivize multiple new generics to enter than if the AG had been present.

How many independent generics are deterred by the presence of the AG also depends on market-specific factors, particularly generic entry costs.

7.4 Discussion on No-AG settlements

No-AG settlement are struck by the brand manufacturer to delay the first generic entrant, resulting in the exclusivity period lasting longer and consumers paying high prices over this extended time. See Section 4.3 and FTC (2011) for an in-depth discussion.[34] Due to lack of data, we do not seek to model no-AG settlements and their welfare implications. Instead, we offer a brief discussion on the implications of no-AG settlements through the lens of our model.

Generally, no-AG settlement means less competition at the beginning of loss-of-exclusivity – the generics entering in the first few quarters do not face an additional competitor in the authorized generic, resulting in higher prices.

However, the negative effects of a no-AG settlement can be partially negated by higher generic entry – no AG means one less competitor in the market, leading to more generics deciding to enter the molecule-formulation. The exact effect of a no-AG settlement on generic entry can therefore depend on how early the generic manufacturers find out that such a deal is in place. Let us suppose that the market conditions were favorable for the AG to be released in the absence of a no-AG settlement. If generics were expecting the AG to enter, then some of them would not apply for ANDA to that market. However if then a no-AG deal is struck, there will be fewer generics, at least in the short term. Potential entrants realize the AG will not in fact be released and file for ANDA approval, resulting in delayed generic entry than otherwise. If generics realize no-AG settlement has been struck early enough, then they can apply for ANDA earlier, resulting in earlier product launch. One policy proposal could therefore be that the brand manufacturer needs to immediately announce when a no-AG settlement has been signed, which will allow additional generics to start applying for entry.

[34]The FTC also filed an amicus brief regarding no-AG settlements, see https://www.ftc.gov/legal-library/browse/amicus-briefs/re-effexor-xr-antitrust-litigation-0

The above discussion highlights the interesting dynamics behind no-AG settlement and their welfare discussions. Furthermore, no-AG settlement may be more complex than described above; see Bokhari et al. (2020) for a discussion. In short, this is potentially an exciting avenue for future research with antitrust implications.

8 Conclusion

In the pharmaceutical industry, branded drug manufacturers can compete with generics by releasing an Authorized Generic, which is identical to the branded drug but without the brand label attached. Using quarterly drug sales and revenue data on US for 2004-2016, we estimate a structural model of pharmaceutical demand, generic entry, AG release, and pricing. We first set up and estimate a discrete choice model of demand where the consumer first chooses a group (brand, nonbrand, or outside option) then chooses a product within the group. Next we build a two-stage supply model of static generic entry followed by a dynamic AG release timing game. The model is solved using backward induction to recover equilibrium generic entrants and conditional choice probabilities of AG release. We estimate this model to recover supply-side parameters.

We simulate market equilibrium under different environments. First, we show how deterrence works in this setting - generics deter through being able to file for entry earlier (early-mover advantage) while AGs can deter because they can launch the product earlier. Next, we find that a faster approval rate leads to weakly fewer generic entrants. This is because greater competition upon launch offsets the advantage from an early launch. Finally, we show that a ban on authorized generics leads to weakly more generic entrants and ambiguous effect on overall prices. The latter happens because while there is increased price competition from weakly more generic entrants in the later stages of the market, it is counteracted by lack of an additional competitor through the AG in the early stages of the market. The exact effect depends on how many generics is being deterred by the AG; for a large number of market specifications, we find that the AG may deter multiple generics.

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A Model solution and cost estimation

A.1 Solving the full model

We solve the model by backward induction and recover the equilibrium number of generic entrants and CCPs of authorized generics.

Note that the AG manufacturer knows the total generics approved in t = 2, i.e. $\mathscr{N}_2$, but can only form probabilistic beliefs about future generic launches. In contrast, generic manufacturers only know the equilibrium generic entrants n_e^*, not $\mathscr{N}_2$.

However, the CCP of the AG varies based on the exact value of $\mathscr{N}_2$, since they suggest different probability distributions over future generic launches. That is, the CCP of AG varies by $\mathscr{N}_2$.

The CCP of the AG is used to calculate the value function of the generics in stage 1. While the generics don't know the exact value of $\mathscr{N}_2$, they can form probabilistic beliefs about it using the FDA approval rates.

Thus, for the model solution, first we solve for the CCP of the AG and construct the corresponding generic value function for each $\mathscr{N}_2$. Then each generic value function (at the initial state s_1) is weighted by the probability of the corresponding $\mathscr{N}_2$ being realized. The resulting weighted generic value function is then used for the first stage.

The solution method is as follows:

1. For a given number of equilibrium generic entrants $\bar{n}$:
   (a) For each $\mathscr{N}_2$ between 0 to $\bar{n}$:
      i. Calculate the authorized generic's state transition matrices as determined by $\bar{n}$, T, FDA approval rates, and $\mathscr{N}_2$.
      ii. Solve for conditional choice probabilities (CCPs) of authorized generic by backward induction from terminal period T. The CCP gives the probability of AG being released in every state of the world from $t = 1$ to T.
      iii. Use the CCP, (14), T, and FDA approval rates to calculate the state transition matrices for generics.

iv. These state transition matrices are combined with per-period profits to construct value functions of generics.

v. Separately, calculate the probability of this $\mathscr{N}_2$ being realized using data on FDA approval rates.

(b) For the initial state s_1, find the generic value function by taking the probability-weighted average of the generic value functions for each $\mathscr{N}_2$ at the state s_1. The probability of each $\mathscr{N}_2$ is being used as weights in this weighted average.

(c) Denote this value function as $V(\bar{n})$ (other arguments suppressed).

2. Repeat the above steps for $\bar{n} = 1, \ldots N$ where N is a large integer.

3. The equilibrium number of generics n^* is such that $V(n^*) \geq \kappa_m^g$ and $V(n^*+1) < \kappa_m^g$. The value function is decreasing in $\bar{n}$.

4. For this n^* re-solve the model to get the equilibrium CCPs of authorized generics conditional on n^* and every possible $\mathscr{N}_2$.

Note that during simulations for a given market, we run many simulations for each possible $\mathscr{N}_2$, then report probability-weighted averages of each summary statistic (such as AG release fraction, AG release time, mean prices, etc.). The probability of each $\mathscr{N}_2$ is being used as weights in this weighted average. We rescale probabilities if some $\mathscr{N}_2$ see no AG released.

A.2 AG entry cost estimation

For AG cost estimation, we can skip a large chunk of the full solution method. From the data we observe n_{em}^* for each market m as well as $\mathscr{N}_{m2}$.

For a given value of parameters:

1. We solve for conditional choice probabilities (CCPs) of authorized generic by backward induction from terminal period T. The CCP gives the probability of AG being released in every state of the world from $t = 1$ to T.

2. The choice probabilities are used to calculate the likelihood.

This process is repeated for different trial values of parameters until the objective function is minimized.

A.3 Estimation of marginal costs

Fix some trial value of generic entry cost κ_g across all molecule-formulations. We try κ_g = \$3 million and \$15 million, which are the publicly reported lower and upper bounds on generic's sunk entry costs. (Intuitively, the \$3 and \$15 million provide upper and lower bounds on the marginal cost scaling parameter ϑ respectively, as we explain below.) For a given value of κ_g, we pick a grid of values for ϑ at 0.01 intervals between 0 and 1. For each value on the grid, solve for the equilibrium number of generic entrants.

The objective function is the sum of squared differences between the observed generic entrants in each market and the predicted entrants. One issue is that the model cannot predict less than zero entrants for a market (i.e. the outcome variable is censored at 0) and so the objective function may be minimized by predicting zero entrants for some markets. To tackle this problem, the objective function also includes a "censoring penalty", where if a market is predicted to have zero generic entrants, it is assigned the censoring penalty value instead of the difference between observed and predicted entrants. This is to prevent the estimation from assigning zero entrants to lots of markets to minimize the objective function. In our estimation, we try censoring penalties of 3 and 4 and obtain similar results. We find the objective function to be convex over our grid of ϑ values.

Returning to the estimation, for a given κ_g, we calculate the objective function for every value on the grid. We pick the value that minimizes the objective function. This gives us the ϑ for that κ_g. We repeat this process for the different κ_g mentioned previously.

B Derivation of Product-Level Market Share

We now derive the expression for product-level market share. We follow the notation of Conlon and Gortmaker (2020). Our derivation follows similar derivations in Berry et al. (1995), Conlon and Gortmaker (2020), Nevo (2000), and other related papers. Throughout the derivation, the time subscript is suppressed for clarity.

We can rewrite Stage 1 in terms of mean utilities and match values following Berry et al. (1995).

$$
\begin{aligned}
u_{ig} = \ & \gamma_m^{(1)} + \lambda_t^{(1)} + \alpha_i^{(1)} ln\, p_g + \beta_{1i}^{(1)} \text{brand}_g + \beta_2^{(1)} brand_g \cdot \text{time-since-loe} + \xi_g^{(1)} + \varepsilon_{ig} \\
= \ & \gamma_m^{(1)} + \lambda_t^{(1)} + \alpha^{(1)} ln\, p_{gt} + \beta_{1i}^{(1)} \text{brand}_g + \beta_2^{(1)} brand_g \cdot \text{time-since-loe} + \xi_g^{(1)} + \\
& \quad \sigma_\alpha \nu_{\alpha,i} ln\, p_g + \sigma_\beta \nu_{\beta,i} \text{brand}_g + \varepsilon_{ig} \\
= \ & \delta_g^{(1)} + \mu_{ig} + \varepsilon_{ig}
\end{aligned}
$$

where the mean utility of group g, $\delta_g^{(1)}$, is given by:

$$
\delta_g^{(1)} = \gamma_m^{(1)} + \lambda_t^{(1)} + \alpha^{(1)} ln\, p_g + \beta_1^{(1)} \text{brand}_g + \beta_2^{(1)} brand_g \cdot \text{time-since-loe} + \xi_g^{(1)}
$$

and the match value of individual i with group g, μ_{ig}, is given by:

$$
\mu_{ig} = \sigma_\alpha \nu_{\alpha,i} ln\, p_g + \sigma_\beta \nu_{\beta,i} \text{brand}_g
$$

Let the idiosyncratic shocks ε_{ig} and ε_{ij} follow a Type-1 Extreme Value distribution. Then the probability of consumer i choosing group g is given as:

$$
s_{ig} = \frac{e^{\delta_g^{(1)} + \mu_{ig}}}{\sum_l e^{\delta_l^{(1)} + \mu_{il}}}
$$

For individual i, let μ_i be the vector of her match-values for all groups in the market, i.e. if both brand and nonbrand drugs are in market, μ_i is 3×1. Similarly, let δ_i be the vector of her mean-values for all products in the market. Since there are many consumers with heterogeneous preferences (reflected in the random coefficients $\alpha_i^{(1)}$ and $\beta_i^{(1)}$), the market share of group g is given by integrating over the distribution of random coefficients:

$$
s_g = \int \frac{e^{\delta_g^{(1)} + \mu_{ig}}}{\sum_l e^{\delta_l^{(1)} + \mu_{il}}} f(\mu_i | \tilde{\theta}_2) d\mu_i
$$

where $\theta_2 = (\sigma_\alpha, \sigma_\beta)$

Conditional on choosing nonbrand group, the probability of a consumer choosing product j from nonbrand group is given by:

$$s_{j|g} = \frac{e^{\delta_j^{(2)}}}{\sum_k e^{\delta_k^{(2)}}}$$

where $\delta_k^{(2)} = \gamma_{m(k)}^{(2)} + \beta^{(2)}\text{AG}_k + \alpha^{(2)} ln(p_k) + \xi_k^{(2)}$.

Stage 1 gives the expression for the market share of a group, while Stage 2 gives the expression for the market share of a product conditional on a group. Therefore, we can express the market share of a single good in terms of group market share and market share of good conditional on choice of group.

$$d_{ij} = \begin{cases} 1 & \text{if } U_{ij} > U_{ik} \text{ for all } k \neq j, \\ 0 & \text{otherwise.} \end{cases}$$

Therefore we can express the market share of a single good in terms of group market share and market share of good conditional on choice of group:

$$\begin{aligned}
s_j =& \int d_{ij}(\delta + \mu_i) d\mu_i d\varepsilon_i \\
=& \int \frac{e^{\delta_g + \mu_{ig}}}{\sum_l e^{\delta_l + \mu_{il}}} \frac{e^{\delta_j + \mu_{ij}}}{\sum_k e^{\delta_k + \mu_{ik}}} f(\mu_i|\tilde{\theta}_2) d\mu_i \\
=& \int \frac{e^{\delta_g + \mu_{ig}}}{\sum_l e^{\delta_l + \mu_{il}}} \frac{e^{\delta_j}}{\sum_k e^{\delta_k}} f(\mu_i|\tilde{\theta}_2) d\mu_i \\
=& \left[\int \frac{e^{\delta_g + \mu_{ig}}}{\sum_k e^{\delta_l + \mu_{il}}} f(\mu_i|\tilde{\theta}_2) d\mu_i \right] \frac{e^{\delta_j}}{\sum_k e^{\delta_k}} \\
=& s_{g(j)} s_{j|g(j)}
\end{aligned}$$

Therefore,

$$s_j = s_{g(j)} s_{j|g(j)}$$

C Counterfactual results

For implementing the counterfactuals, we divide the 241 molecule-formulations into "small", "medium", and "large" markets. The "small" markets are assigned generic entry cost of \$1 million, "medium" markets are assigned per-period fixed cost of and generic entry cost of \$3 million, and "large" markets are assigned per-period fixed cost of and generic entry cost of \$15 million .

Now we explain how we classify markets into these 3 groups. The general problem is that for the lowest entry cost of \$3 million, a number of molecule formulations predict 0 generic entrants or more than 22 generic entrants (the highest number observed in our dataset).

As a result, for each molecule-formulation, we try the three generic entry costs mentioned previously. A molecule-formulation is assigned to the group with the largest costs such that the model predicts positive generic entrants less than 22 and more than zero.

For 41 of the 241 molecule-formulations our model does not predict positive generic entrants, and we drop those markets from our counterfactuals. In the end, we have 31 molecule-formulations in the "small" markets, 163 molecule-formulations in the "medium" markets, and 6 molecule-formulations in the "large" markets.

Note that we think of counterfactuals under different market specifications as mainly checking how our model predictions vary for different realistic parameter values.

A note on terminology: throughout this section "Early periods" refers to first 4 quarters, and "Later periods" refers to last 4 quarters (the terminal period is quarter 32).

C.1 Early-move vs early-launch

Table 8: Distribution of outcomes across market-specifications.

Percentiles	**AG release fraction**
10%	0.0
20%	0.0
30%	0.0
40%	0.0
50%	0.0
60%	0.0
70%	0.0
80%	0.0
90%	0.0

Notes: For this counterfactual we fix the following: i) Generics can launch immediately upon loss-of-exclusivity (at $t = 2$), so that the AG no longer enjoys a launch timing advantage over generics; ii) AG has a low logit scaling parameter so that the AG can launch immediately; iii) AGs have the same entry cost as generics, so that the AG no longer has a cost advantage; iv) AG and generics face the same demand primitives, so that neither enjoy any demand-side advantage over the other. We find that AG almost never launches.

Table 9: Distribution of outcomes across market-specifications.

Percentiles	**Difference in generics**
10%	-1.0
20%	-1.0
30%	-1.0
40%	0.0
50%	0.0
60%	0.0
70%	0.0
80%	0.0
90%	0.0

Notes: We vary the AG's early-launch advantage to see how it affects generic entry. First we run simulations where the AG has low logit scaling parameter, which allows it to launch very early without any delays. Next we run simulations where the AG has a high logit scaling parameter, which means it launches much later due to delays. "Difference in generics" is the equilibrium generic entrants when the AG launches early minus the equilibrium generic entrants when the AG launches late. We limit to market-specifications where AG launches at least once across all simulations for both low and high logit scaling parameters. We find that across most market-specifications, the AG can deter more generics when it can launch earlier.

C.2 Faster generic approval rates

Table 10: Distribution of outcomes across market-specifications with estimated versus immediate generic approval. The AG is prevented from entering for expositional clarity.

Percentiles	Difference in no. of generics	Difference in generics price (early periods)	Difference in generics price (later periods)
10%	0.0	0.115	-3.07
20%	0.0	0.256	-0.742
30%	0.0	0.387	-0.414
40%	1.0	0.619	-0.212
50%	1.0	0.891	-0.122
60%	1.0	1.463	-0.044
70%	1.0	3.337	0.0
80%	1.0	7.288	0.0
90%	2.0	23.612	0.0

Notes: "Difference in no. of generics" is the equilibrium generic entrants under estimated generic approval rates minus the equilibrium generic entrants under immediate generic approval. Other variables similarly defined. We find that faster generic approval rates lead to fewer generics entering the market.

Table 11: Distribution of outcomes across market-specifications with estimated versus immediate generic approval.

Percentiles	**Difference in no. of generics**	**Difference in generics price (early periods)**	**Difference in generics price (later periods)**
10%	0.0	0.044	-1.486
20%	0.0	0.108	-0.216
30%	0.0	0.253	-0.086
40%	0.0	0.368	-0.008
50%	0.0	0.607	0.0
60%	0.0	0.98	0.0
70%	1.0	2.192	0.0
80%	1.0	6.113	0.0
90%	1.0	20.067	0.0

Notes: "Difference in no. of generics" is the equilibrium generic entrants under estimated generic approval rates minus the equilibrium generic entrants under immediate generic approval. Other variables similarly defined. We find that faster generic approval rates lead to fewer generics entering the market.

C.3 Ban on AG

Table 12: Distribution of outcomes across market-specifications.

Percentiles	**Difference in no. of generics**	**Difference in generics price (early periods)**	**Difference in generics price (later periods)**
10%	-2.0	-0.54	-0.0
20%	-2.0	-0.271	0.0
30%	-2.0	-0.171	0.0
40%	-2.0	-0.088	0.018
50%	-2.0	-0.056	0.049
60%	-2.0	-0.025	0.118
70%	-1.0	-0.016	0.231
80%	-1.0	-0.003	0.411
90%	-1.0	0.112	1.066

Notes: "Difference in no. of generics" is the equilibrium generic entrants when AG is allowed minus equilibrium generic entrants when AG is banned. Other variables similarly defined. We find that banning AG generally leads to more generic entrants. We assume that in the absence of a ban, the AG would have drawn a trade shock and therefore could have chosen to enter.

Table 13: Distribution of outcomes across market-specifications.

Percentiles	**Difference in no. of generics**	**Difference in generics price (early periods)**	**Difference in generics price (later periods)**
10%	-2.0	-4.691	0.0
20%	-1.0	-1.229	0.0
30%	-1.0	-0.494	0.024
40%	-1.0	-0.282	0.066
50%	-1.0	-0.203	0.121
60%	-1.0	-0.111	0.2
70%	-1.0	-0.068	0.309
80%	0.0	-0.04	0.675
90%	0.0	-0.02	1.892

Notes: "Difference in no. of generics" is the equilibrium generic entrants when AG is allowed minus equilibrium generic entrants when AG is banned for the first 2 quarters. Other variables similarly defined.

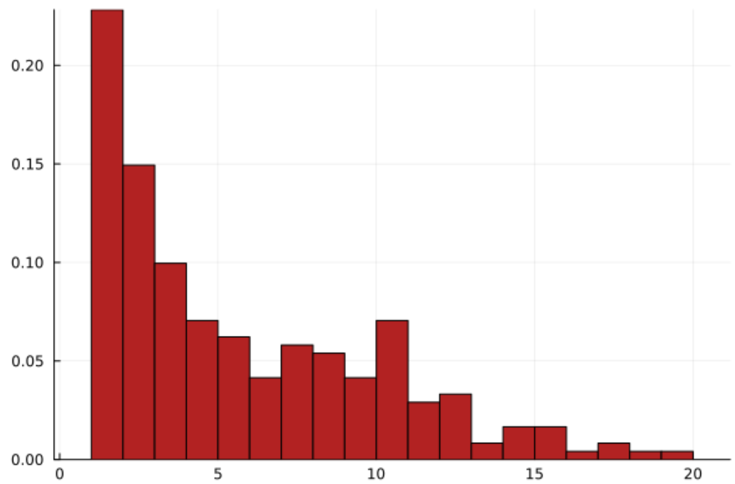
D Additional estimation results

E Additional summary statistics

Delay	Count
0.0	58
1.0	10
2.0	6
3.0	2
4.0	2
5.0	3
6.0	2
8.0	1
9.0	1
10.0	1
11.0	1
12.0	2
14.0	2
15.0	2
17.0	1
19.0	2
20.0	2
21.0	1
22.0	2
27.0	1
38.0	1
41.0	1

Table 14: AG Delay table. This shows, of the 110 AGs released, how many quarters after loss-of-exclusivity they were released. “Delay” is the number of quarters after loss-of-exclusivity. “Count” is the number of AGs released for the corresponding quarters since loss-of-exclusivity.

Figure 5: Histogram of total generic entry by molecule-formulation



Notes: This distribution has a 25th percentile of 2, median of 4, 75th percentile of 8, and 90th percentile of 11. There are 241 molecule-formulations.

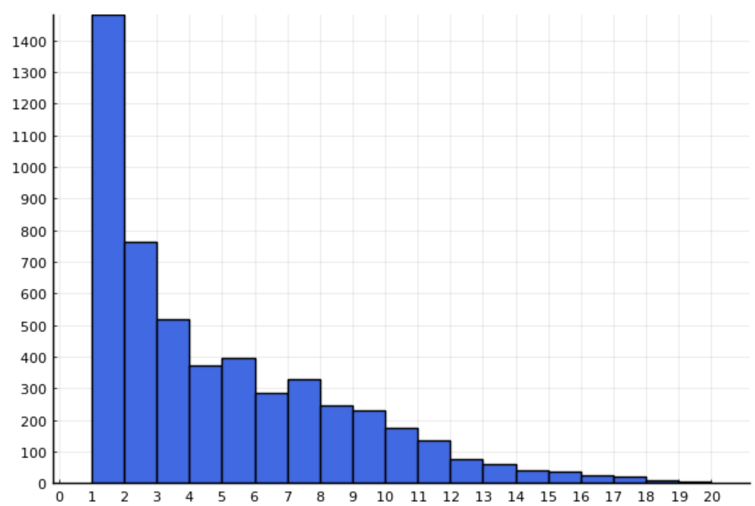
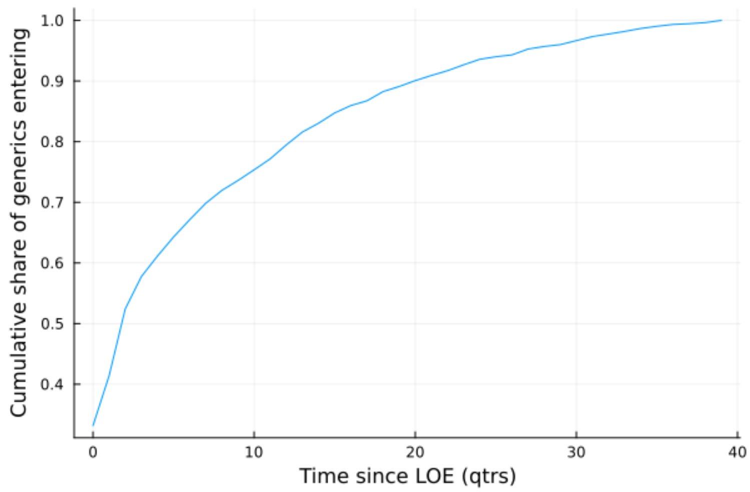


Figure 6: Histogram of count of generics present by molecule-formulation-quarter.

Figure 7: Generic entry rate



Notes:

Table 15

Percentiles	10%	25%	50%	75%	90%
Generic revenue	0.01	0.13	0.79	3.13	9.2
AG revenue	0.1	0.63	3.46	8.67	24.98
Brand revenue	0.18	0.87	4.41	22.08	99.1
Mean brand revenue	0.35	1.2	4.94	19.84	67.81
Mean generic revenue	0.29	0.82	2.35	5.36	12.33
Mean AG revenue	0.32	1.27	4.32	12.15	25.19
Mean brand within molform mktshare	3.51	6.21	15.61	34.92	69.76
Mean generic within molform mktshare	7.96	11.05	18.54	31.91	51.59
Mean AG within molform mktshare	4.19	11.57	22.04	36.01	49.7
Generic total lifetime revenue by product	0.26	1.73	11.06	54.78	184.7
AG total lifetime revenue by product	3.97	10.99	54.45	201.78	444.5

Notes:

F Miscellaneous

F.1 Comparison with Fowler et al. (2023)

One difference between Fowler et al. (2023) and our findings is they find AG incidence is just 6% of the markets, while we find nearly 44%. We believe this is mainly because they define product and possible markets for AG release ("at risk markets") more broadly than us. In particular, differences in AG entry rate arise due to:

1. Definition of product: We consider a product market to be "molecule-formulation", where formulation is either oral, injectable, or everything else. Fowler et al. (2023) define a product to vary by strength and more detailed dosage forms. As a result, their number of markets is much larger than ours.

2. At-risk markets: We consider a market to potentially see AG entry (at-risk market) if we find that it has seen at least one generic entry (and that first generic entry happened after 2004). Instead, Fowler et al. (2023) considers at-risk products to be "number of products approved during or after 1985 that had not been withdrawn or discontinued and that still had patent or exclusivity protection in each year from 2010 to 2019". This is a very broad definition. Since AG release is only considered when a generic enters, our definition is consistent with our reesarch question.

3. Undercounting AGs: Fowler et al. (2023) mentions that for data issues "... led to conservative estimates of the incidence of authorized generic entry"

4. Difference in yearly coverage: We look at 2004-2016, while they consider 2010-2019.

Other findings from Fowler et al. (2023) match our paper closely.

F.2 Comparison of generic approval rates with Starc and Wollmann (2023)

Another paper that measures FDA approval rates of generics is Starc and Wollmann (2023). Their setting and data is different from ours. Starc and Wollmann (2023) studies generic application and entry for markets where the loss-of-exclusivity has happened well in the past. They also have data on when the generic filed their ANDA, and so they can exactly measure the time between filing and approval. They report the distribution of "regulatory delays" in years and report it in a graph; numerically they find the following distribution:

- 1 year: 0.1
- 2 year: 0.34
- 3 year: 0.34
- 4 year: 0.15
- 5 year: 0.07
- 6 year: 0.04

(These figures are read off a graph and so aren't exact.)

Compared to our estimates, they find much slower ANDA approval rates. We are not sure why that is. One reason could be that we are modeling markets which face loss-of-exclusivity. For some of these markets, the loss-of-exclusivity is anticipated, and so generics may start filing in advance. As a result, they arrive on the market much sooner compared to the setting in Starc and Wollmann (2023), where generics notice a cartel formation/high prices due to cartel formation and then apply for ANDA approval.

As a result we think that while Starc and Wollmann (2023) calculates approval rates that are correct for their setting, they might be too slow for our setting.

One weakness to our approach is that we do not have data on when generics file their ANDAs, and so we cannot directly measure the time between filing and approval. Instead what we do is calculate the rate at which generics launch in the market and use it to recover the approval rate.

An alternative is we can re-estimate our model and counterfactuals using the ANDA approval rates found in Starc and Wollmann (2023).

G Useful quotes from FTC (2011)

G.1 Quotes on how the brand manufacturer tightly controls the timing of AGs

- (Pg 73)“Early launch, however, “cannibalizes” sales of the brand-name product, starting their decline before ANDA-generic entry has occurred. Brand-name firms have chosen to tightly control the timing of AG launch to avoid unnecessary loss of brand sales and other disadvantages. Launch even a short time before confirmation of ANDA-generic entry is disfavored, and AG supply agreements often include strict penalty provisions or financial terms that enforce launch timing.”

- (Pg 73 footnote) “See CD, Nov. 4, 2004 (“financial risk to brand [of early launch is] significant” because brand erosion cannot be “made up by benefit of 2 week early launch”; that risk is exacerbated if the first generic entrant is delayed for any reason, because the AG would cannibalize the brand during the delay); CD, Feb. 2, 2005 (“Do not put Rx brand at risk (i.e., do not launch in advance of a generic, but can have contracts in place and inventory at wholesalers”)) (emphasis original); CD, Jan. 5, 2007 (“[generic subsidiary] and [brand-name company] agree it is not financially viable to launch an AG prior to launch of other AB rated generics in the US”); CD, Mar. 31, 2004 (disadvantages of launching the AG early include “Uncertainty over ultimate LOE date [that] could cause unnecessary brand erosion; Additional incentive for earlier ‘at risk’ launches by generics; Potential public relations issue; Legislative Risks”).”

- “(Pg 21) The agreements also establish the timing of AG launch. Many agreements provide for launch only after ANDA-generic entry or the date of expected ANDA-generic launch. To ensure that brand sales are not eroded before ANDA-generic competition begins, penalty provisions often apply if the AG marketer launches before generic entry”

G.2 Quotes on outsourcing AG production

- (Pg 77)"Profit-splits, i.e., the percentage of profits that the AG distributor must pay to the brand, are also consistent with the brand's interest in the revenue stream arising from an AG. Apart from contexts where AG marketing rights derive from patent litigation settlements, marketing agreements generally require AG distributors to pay the brand a large percentage of profits on the AG, typically from 50-92 percent."

- (Pg 85) "At the same time, some documents also suggest that AG distribution opportunities for generic firms even as early as 2004 were becoming more difficult to obtain, thereby prompting some to re-visit their AG strategies.30 As one generic firm document explains: AGx opportunities and expected returns have both been substantially reduced in recent months, due to the decision of various brand companies to market more of their products through their own AGx subsidiaries, and increased competition among generic companies for available AGx deals...

  Revenue and earnings shortfalls by many publicly-traded generic companies due to severe pricing competition has made AGs one of the only ways to maintain growth in the short term; The resulting intense competitive bidding for remaining opportunities has made brand companies realize they are in a position to retain most of the economic value of authorized generic versions of their intellectual property; Historically, generic partners received 20-35% of the economics, but this has now been driven down to 10% or less; The reduced value of AG deals for generic companies will hurt the business models of some companies..."

- "(Pg 23-24) A number of terms affect potential profitability: • Pricing: Neither settlement nor non-settlement agreements restrict the generic company's authority in setting the price of AGs. Indeed, they frequently affirm that authority. • Transfer Price: The transfer or supply price establishes the amount the generic company pays the brand-name company for the AG. In some agreements, this price is related by express terms to the cost of manufacturing the AG.35 • Profit-Splits: Most agreements provide a profit-split, which governs the distribution of profits between the brand and generic companies from sales of the AG. Many non-settlement agreements require the AG marketer to pay the brand a large percentage of profits on the AG.36

Profit-splits have implications for generic firms' incentives: although AG distribution agreements can give a generic company that was not the first filer the benefit of being able to market during the 180-day exclusivity period, first-filing status provides the possibility of exclusivity unburdened by a profit-split, albeit with the likely expense of patent litigation."

G.3 Quotes on how AGs are usually not aimed at 180-day

- "(Pg 27) Although drug companies marketed many AGs during 180-day exclusivity, they marketed more during non-exclusive periods, and often for drugs that had not been subject to a patent challenge. "

- "(Pg 32) Taken together, the exclusivity and patent certification data in Figures 2-8 and 2-10 indicate that during the period when AGs became common, undermining 180-day exclusivity was unlikely to have been the primary rationale for many AG drugs. This does not preclude a disincentive effect or rationale for those AGs that were marketed during 180-day exclusivity periods, but it suggests that different or additional explanations for AG marketing also should be considered. "

- "(Pg 32) Nonetheless, AGs marketed during exclusivity were in the minority. More AGs were marketed after exclusivity or when no exclusivity had occurred; many of the latter drugs had not been subject to a patent challenge. The rate of AG launch increased as pre-entry brand sales increased, and within most sales levels was similar for ANDA-generic entry with and without exclusivity; there was no consistent pattern in the relative magnitude of rates of AG entry for generic entry with and without exclusivity. These findings suggest that, on average, the anticipated sales level of the brand-name drug has had greater bearing on decisions to launch AGs than whether ANDA-generic entry occurred via exclusivity. The findings, however, do not preclude a disincentive rationale or effect for AGs that were marketed during exclusivity."